and V is the volume (dm^3).
Thus, D is expressed in the units h^{-1}.

The net change in cell concentration over a time period may be expressed as:

$$dx/dt = \text{growth} - \text{output}$$

or

$$dx/dt = \mu x - Dx. \tag{2.10}$$

Under steady-state conditions the cell concentration remains constant, thus $dx/dt = 0$ and:

$$\mu x = Dx \tag{2.11}$$

and

$$\mu = D. \tag{2.12}$$

Thus, under steady-state conditions the specific growth rate is controlled by the dilution rate, which is an experimental variable. It will be recalled that under batch culture conditions an organism will grow at its maximum specific growth rate and, therefore, it is obvious that a continuous culture may be operated only at dilution rates below the maximum specific growth rate. Thus, within certain limits, the dilution rate may be used to control the growth rate of the culture.

The growth of the cells in a continuous culture of this type is controlled by the availability of the growth limiting chemical component of the medium and, thus, the system is described as a chemostat. The mechanism underlying the controlling effect of the dilution rate is essentially the relationship expressed in equation (2.5), demonstrated by Monod in 1942:

$$\mu = \mu_{max} s/(K_s + s).$$

At steady state, $\mu = D$, and, therefore,

$$D = \mu_{max}\bar{s}/(K_s + \bar{s})$$

where $\bar{s}$ is the steady-state concentration of substrate in the chemostat, and

$$\bar{s} = K_s D/(\mu_{max} - D). \tag{2.13}$$

Equation (2.13) predicts that the substrate concentration is determined by the dilution rate. In effect, this occurs by growth of the cells depleting the substrate to a concentration that supports the growth rate equal to the dilution rate. If substrate is depleted below the level that supports the growth rate dictated by the dilution rate, the following sequence of events takes place:

(i) The growth rate of the cells will be less than the dilution rate and they will be washed out of the vessel at a rate greater than they are being produced, resulting in a decrease in biomass concentration.
(ii) The substrate concentration in the vessel will rise because fewer cells are left in the vessel to consume it.
(iii) The increased substrate concentration in the vessel will result in the cells growing at a rate greater than the dilution rate and biomass concentration will increase.
(iv) The steady state will be re-established.

Thus, a chemostat is a nutrient-limited self-balancing culture system which may be maintained in a steady state over a wide range of sub-maximum specific growth rates.

The concentration of cells in the chemostat at steady state is described by the equation:

$$\bar{x} = Y(S_R - \bar{s}) \tag{2.14}$$

where $\bar{x}$ is the steady-state cell concentration in the chemostat.

By combining equations (2.13) and (2.14), then:

$$\bar{x} = Y[S_R - \{K_s D/(\mu_{max} - D)\}]. \tag{2.15}$$

Thus, the biomass concentration at steady state is determined by the operational variables, S_R and D. If S_R is increased, $\bar{x}$ will increase but $\bar{s}$, the residual substrate concentration in the chemostat, will remain the same. If D is increased, μ will increase ($\mu = D$) and the residual substrate at the new steady state would have increased to support the elevated growth rate; thus, less substrate will be available to be converted into biomass, resulting in a lower steady state value.

An alternative type of continuous culture to the chemostat is the turbidostat, where the concentration of cells in the culture is kept constant by controlling the flow of medium such that the turbidity of the culture is kept within certain, narrow limits. This may be achieved by monitoring the biomass with a photoelectric cell and feeding the signal to a pump supplying medium to the culture such that the pump is switched on if the biomass exceeds the set point and is switched off if the biomass falls below the set point. Systems other than turbidity may be used to monitor the biomass concentration, such as CO_2 concentration or pH in which case it would be more correct to term the culture a biostat. The chemostat is the more commonly used system because it has the advantage over the biostat of not requiring complex control systems to maintain a steady state. However, the biostat may be advantageous in continuous enrichment culture in avoiding the total washout of the culture in its early stages and this aspect is discussed in Chapter 3.

The kinetic characteristics of an organism (and, therefore, its behaviour in a chemostat) are described by the numerical values of the constants Y, μ_{max} and K_s. The value of Y affects the steady-state biomass concentration; the value of μ_{max} affects the maximum dilution rate that may be employed and the value of K_s affects the residual substrate concentration (and, hence, the biomass concentration) and also the maximum dilution rate that may be used. Figure 2.4 illustrates the continuous culture behaviour of a hypothetical bacterium with a low K_s value for the limiting substrate, compared with the initial limiting substrate concentration. With increasing dilution rate, the residual substrate concentration increases only slightly until D approaches μ_{max} when s increases significantly. The dilution rate at which x equals zero (that is, the cells have been washed out of the system) is termed the critical dilution rate (D_{crit}) and is given by the equation:

$$D_{crit} = \mu_{max} S_R/(K_s + S_R). \qquad (2.16)$$

Thus, D_{crit} is affected by the constants, μ_{max} and K_s, and the variable, S_R; the larger S_R the closer is D_{crit} to μ_{max}. However, μ_{max} cannot be achieved in a simple steady state chemostat because substrate limited conditions must always prevail.

Figure 2.5 illustrates the continuous culture behaviour of a hypothetical bacterium with a high K_s for the limiting substrate compared with the initial limiting substrate concentration. With increasing dilution rate, the residual substrate concentration increases significantly to support the increased growth rate. Thus, there is a gradual increase in s and a decrease in x as D approaches D_{crit}. Figure 2.6 illustrates the effect of increasing the initial limiting substrate concentration

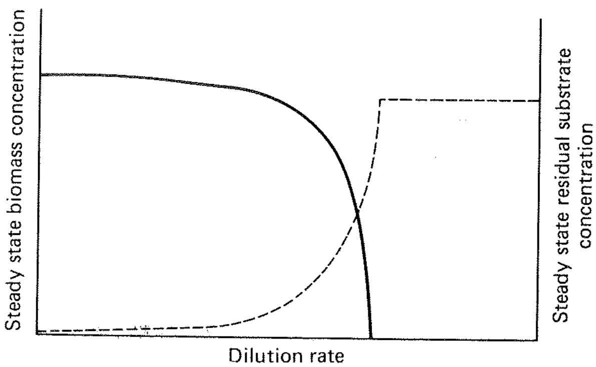

FIG. 2.4. The effects of dilution rate on the steady-state biomass and residual substrate concentrations in a chemostat culture of a micro-organism with a low K_s value for the limiting substrate, compared with the initial substrate concentration.
——— Steady-state biomass concentration.
– – – Steady-state residual substrate concentration.

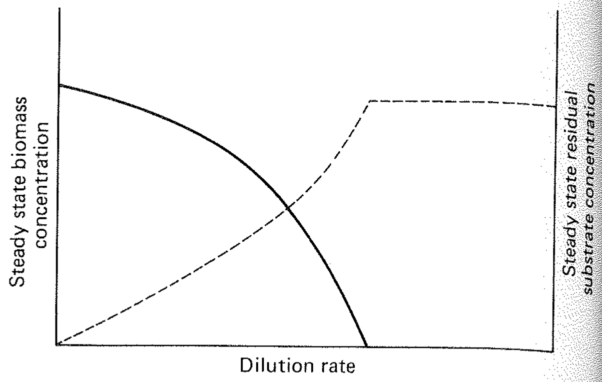

FIG. 2.5. The effect of dilution rate on the steady-state biomass and residual substrate concentrations in a chemostat of a micro-organism with a high K_s value for the limiting substrate, compared with the initial substrate concentration.
——— Steady-state biomass concentration.
– – – Steady-state residual substrate concentration.

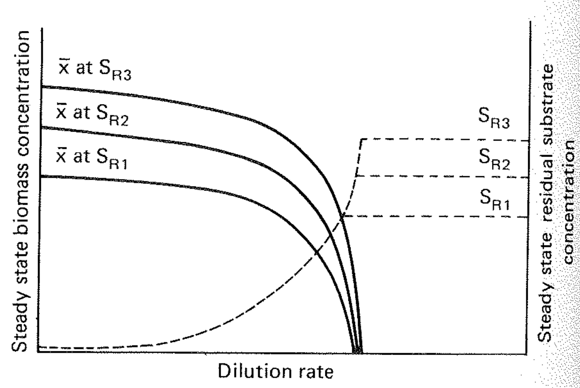

FIG. 2.6. The effect of the increased initial substrate concentration on the steady-state biomass and residual substrate concentrations in a chemostat.
——— Steady-state biomass concentrations.
– – – Steady-state residual substrate concentrations.
S_{R1}, S_{R2} and S_{R3} represent increasing concentrations of the limiting substrate in the feed medium.

on $\bar{x}$ and $\bar{s}$. As S_R is increased, so $\bar{x}$ increases, but the residual substrate concentration is unaffected. Also, D_{crit} increases slightly with an increase in S_R.

The results of chemostat experiments may differ from those predicted by the foregoing theory. The reasons for these deviations may be anomalies associated with the equipment or the theory not predicting the behaviour of the organism under certain circumstances. Practical anomalies include imperfect mixing and wall growth. Imperfect mixing would cause an increase in the degree of heterogeneity in the fermenter with some organisms being subject to nutrient excess whilst others are under severe limitation. This phenomenon is particularly relevant to very low dilution rate systems when the flow of medium is likely to be very intermittent. This problem may be overcome by the use of feed-back systems, as discussed later in this Chapter. Wall growth is another commonly encountered practical difficulty in which the organism adheres to the inner surfaces of the reactor resulting, again, in an increase in heterogeneity. The immobilized cells are not subject to removal from the vessel but will consume substrate resulting in the suspended biomass concentration being lower than predicted. Wall growth may be limited by coating the inner surfaces of the vessel with Teflon.

A frequent observation in carbon and energy limited chemostats is that the biomass concentration at low dilution rates is lower than predicted. This is attributed to the phenomenon of micro-organisms utilizing a greater proportion of substrate for maintenance at low dilution rates. Effectively, the yield factor decreases at low dilution rates. Bull (1974) has reviewed the major causes of deviations from basic chemostat theory.

The basic chemostat may be modified in a number of ways, but the most common modifications are the addition of extra stages (vessels) and the feedback of biomass into the vessel.

Multistage systems

A multistage system is illustrated in Fig. 2.7. The advantage of a multistage chemostat is that different conditions prevail in the separate stages. This may be

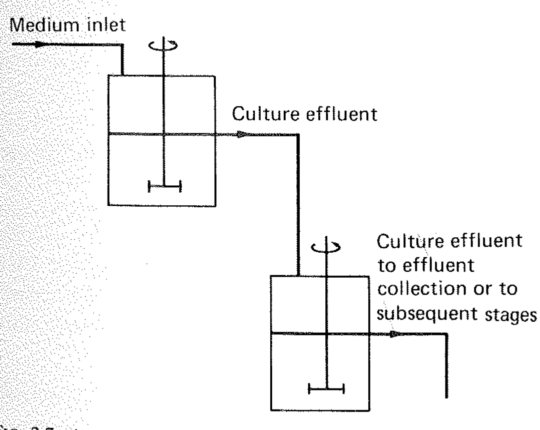

FIG. 2.7. A multistage chemostat.

advantageous in the utilization of multiple carbon sources and in the production of secondary metabolites. Harte and Webb (1967) demonstrated that when *Klebsiella aerogenes* was grown on a mixture of glucose and maltose only the glucose was utilized in the first stage and maltose in the second. Secondary metabolism may occur in the second stage of a dual system in which the second stage acts as a holding tank where the growth rate is much smaller than that in the first stage. The adoption of multistage systems in research and industry has been extremely limited, due to the complexity of the systems. One example of the industrial application of the technique is in continuous brewing which is described in a later section.

Feedback systems

A chemostat incorporating biomass feedback has been modified such that the biomass in the vessel reaches a concentration above that possible in a simple chemostat, that is, greater than $Y(S_R - s)$. Biomass concentration may be achieved by:

(i) Internal feedback. Limiting the exit of biomass from the chemostat such that the biomass in the effluent stream is less concentrated than in the vessel.
(ii) External feedback. Subjecting the effluent stream to a biomass separation process, such as sedimentation or centrifugation, and returning a portion of the concentrated biomass to the growth vessel.

Pirt (1975) gave a full kinetic description of these feedback systems and this account summarizes his analysis.

INTERNAL FEEDBACK

A diagrammatic representation of an internal feedback system is shown in Fig. 2.8a. Effluent is removed from the vessel in two streams, one filtered, resulting in a dilute effluent stream (and, thus, a concentration of biomass in the reactor) and one unfiltered. The proportion of the outflow leaving via the filter and the effectiveness of the filter then determines the degree of feedback. The flow rate of incoming medium is designated F (dm^3 h^{-1}) and the fraction of the outflow which is not filtered is designated c; thus the outflow rate of the unfiltered stream is cF and that of the

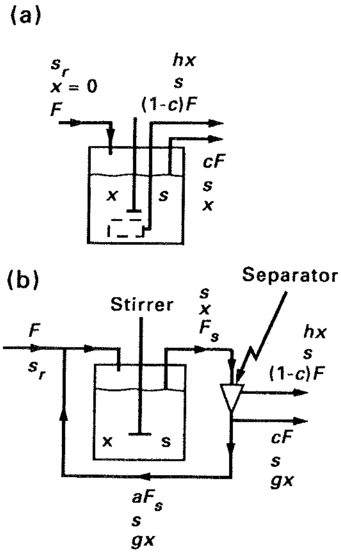

FIG. 2.8. Diagrammatic representations of chemostats with feedback (Pirt, 1975).
(a) Internal feedback.
F = flow rate of incoming medium ($dm^3\ h^{-1}$)
c = fraction of the outflow which is not filtered
x = biomass concentration in the vessel and in theunfiltered stream
hx = biomass concentration in the filtered stream
(b) External feedback.
F = flow rate from the medium reservoir ($dm^3\ h^{-1}$)
F_s = flow rate of the effluent upstream of the separator
x = biomass concentration in the vessel and upstream of the separator
hx = biomass concentration in the dilute stream from the separator
g = factor by which the separator concentrates the biomass
a = proportion of the flow which is fed back to the fermenter
s = substrate concentration in the vessel and effluent lines
S_R = substrate concentration in the medium reservoir

filtered stream is $(1 - c)F$. The concentration of the biomass in the fermenter and in the unfiltered stream is x and the concentration of biomass in the filtered stream is hx. The biomass balance of the system is:

Change in biomass = Growth − Output in unfiltered stream − Output in filtered stream,

which may be expressed as:

$$dx/dt = \mu x - cDx - (1 - c)Dhx \qquad (2.17)$$

or:

$$dx/dt = \mu x - D\{c(1 - h) + h\}x.$$

At steady state $dx/dt = 0$, thus:

$$\mu\bar{x} = D\{c(1 - h) + h\}\bar{x}$$

and

$$\mu = D\{c(1 - h) + h\}.$$

If the term '$c(1 - h) + h$' is represented by 'A', then:

$$\mu = AD. \qquad (2.18)$$

When the filtered effluent stream is cell free then h = 0 and $A = c$. However, if the filter removes very little biomass then h will approach 1, and when $h = 1$ there is no feedback and $A = h$. Thus, the range of values of A is c to h and when feedback occurs A is less than 1, which means that μ is less than D.

The concentration of the growth limiting substrate in the vessel at steady state is then given by:

$$\bar{s} = K_s AD/(\mu_{max} - AD) \qquad (2.19)$$

and the biomass concentration at steady state is given by:

$$\bar{x} = Y/A(S_R - \bar{s}). \qquad (2.20)$$

EXTERNAL FEEDBACK

A diagrammatic representation of an external feedback system is shown in Fig. 2.8b. The effluent from the fermenter is fed through a separator, such as a continuous centrifuge or filter, which produces two effluent streams — a concentrated biomass stream and a dilute one. A fraction of the concentrated stream is then returned to the vessel. The flow rate from the medium reservoir is F ($dm^3\ h^{-1}$); the flow rate of the effluent upstream of the separator is F_s ($dm^3\ h^{-1}$) and the concentration of biomass in the stream (and in the fermenter) is x; a is the proportion of the flow which is fed back to the fermenter and g is the factor by which the separator concentrates the biomass. Biomass balance in the system will be:

Change = growth − output + feedback

or

$$dx/dt = \mu x - F_s x/V + aF_s gx/V. \qquad (2.21)$$

The culture outflow (before separation), F_s, from the chemostat is:

$$F_s = F + aF_s$$

or

$$F_s = F/(1 - a),$$

substituting $F/(1 - a)$ for F_s in equation (2.21) and remembering that $D = F/V$:

$$dx/dt = \mu x - Dx/(1 - a) + agDx/(1 - a). \quad (2.22)$$

If all the cells are returned to the fermenter then biomass will continue to accumulate in the vessel. However, if the feedback is partial then a steady state may be achieved, $dx/dt = 0$ and:

$$\mu = BD \qquad (2.23)$$

where $B = (1 - ag)/(1 - a)$.

The steady-state substrate and biomass concentrations in a fermenter with feedback are then given by the following equations:

$$\bar{s} = BDK_s/(\mu_{max} - D), \tag{2.24}$$

$$\bar{x} = Y/B(S_R - s). \tag{2.25}$$

From the equations describing μ, $\bar{s}$ and $\bar{x}$ (2.18, 2.19, 2.20, 2.23, 2.24, 2.25) in a fermenter with either external or internal feedback it can be appreciated that:

(i) Dilution rate is greater than growth rate.
(ii) Biomass concentration in the vessel is increased.
(iii) The increased biomass concentration results in a decrease in the residual substrate compared with a simple chemostat.
(iv) The maximum output of biomass and products is increased.
(v) Because D is less than μ, the critical dilution rate (the dilution rate at which washout occurs) is increased.

Biomass feedback is applied widely in effluent treatment systems where the advantages of feedback contribute significantly to the process efficiency. The outlet substrate concentration is considerably less and the feedback of biomass may improve stability in effluent treatment systems where mixed substrates of varying concentration are used. The system will also result in increased productivity of microbial products as illustrated by Major and Bull (1989) who reported very high lactic acid productivities in laboratory biomass recycle fermentations. Anaerobic processes are particularly suited to feedback continuous culture because the elevated biomass is not susceptible to oxygen limitation.

Feedback systems seem particularly attractive for animal cell culture where low growth rates and low cell densities limit productivity. A number of internal feedback systems have been developed based on immobilized cells on either hollow fibres or microcarriers (see also Chapter 7) and it is claimed that with the rapid developments in centrifuge design, centrifugal separation and feedback in suspension cultures may be scaled-up (Griffiths, 1992). The potential for continuous animal-cell processes is considered in the next section of this Chapter.

Comparison of batch and continuous culture in industrial processes

BIOMASS PRODUCTIVITY

The productivity of a culture system may be described as the output of biomass per unit time of the fermentation. Thus, the productivity of a batch culture may be represented as:

$$R_{batch} = (x_{max} - x_0)/(t_i - t_{ii}) \tag{2.26}$$

where R is the output of the culture (g biomass $dm^{-3}\ h^{-1}$),
x_{max} is the maximum cell concentration achieved at stationary phase,
x_0 is the initial cell concentration at inoculation,
t_i is the time during which the organism grows at μ_{max},
and t_{ii} is the time during which the organism is not growing at μ_{max} and includes the lag phase, the deceleration phase and the periods of batching, sterilizing and harvesting.

The productivity of a continuous culture may be represented as:

$$R_{cont} = D\bar{x}(1 - t_{iii}/T) \tag{2.27}$$

where R_{cont} is the output of the culture (g biomass $dm^{-3}\ h^{-1}$),
t_{iii} is the time period prior to the establishment of a steady state and includes vessel preparation, sterilization and operation in batch culture prior to continuous operation,
and T is the time period during which steady-state conditions prevail.

The term $D\bar{x}$ increases with increasing dilution rate until it reaches a maximum value, after which any further increase in D results in a decrease in $D\bar{x}$, as illustrated in Fig. 2.9. Thus, maximum productivity of biomass may be achieved by the use of the dilution rate giving the highest value of $D\bar{x}$.

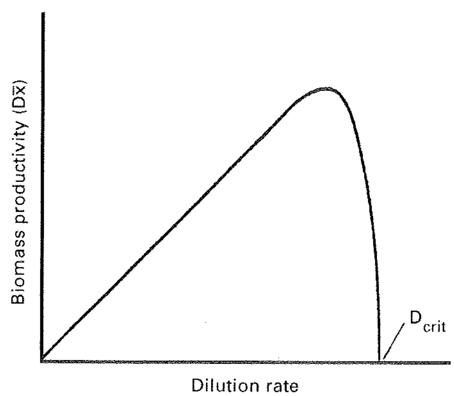

FIG. 2.9. The effect of dilution rate on biomass productivity in steady-state continuous culture.

The output of a batch fermentation described by equation (2.26) is an average over the period of the fermentation and, because the rate of biomass production is dependent on initial biomass ($dx/dt = \mu x$), the vast proportion of the biomass is produced towards the end of the fermentation. Thus, productivity in batch culture is at its maximum only towards the end of the process. For a continuous culture operating at the optimum dilution rate, under steady-state conditions, the productivity will be constant and always maximum. Thus, the productivity of the continuous system must be greater than the batch. A continuous system may be operated for a very long time period (several weeks or months) so that the negative contribution of the unproductive time, t_{iii}, to productivity would be minimal. However, a batch culture may be operated for only a limited time period so that the negative contribution of the time, t_{ii}, would be very significant, especially when it is remembered that the batch culture would have to be re-established many times during the time-course of a continuous run. Thus, the superior productivity of biomass by a continuous culture, compared with a batch culture, is due to the maintenance of maximum output conditions throughout the fermentation and the insignificance of the non-productive period associated with a long-running continuous process.

The steady state achievable in a continuous process also adds to the advantage of improved biomass productivity, as discussed by Hospodka (1966). Cell concentration, substrate concentration, product concentration and toxin concentration should remain constant throughout the fermentation. Thus, once the culture is established the demands of the fermentation, in terms of process control, should be constant. In a batch fermentation, the demands of the culture vary during the fermentation — at the beginning, the oxygen demand is low but towards the end the demand is high, due to the high biomass and the increased viscosity of the broth. Also, the amount of cooling required will increase during the process, as will the degree of pH control. In a continuous process oxygen demand, cooling requirements and pH control should remain constant. Thus, the use of continuous culture should allow for the easier introduction of process automation.

A batch process requires periods of intensive labour during medium preparation, sterilization, batching and harvesting but relatively little during the fermentation itself. However, a continuous process results in a more constant labour demand in that medium is supplied continuously sterilized (see Chapter 5), the product is continuously extracted and the relative time spent on equipment preparation and sterilization is very small.

The argument against continuous biomass processes is that the duration of a continuous fermentation is very much longer than a batch one so that there is a greater probability of a contaminating organism entering the continuous process and a greater probability of equipment failure. However, problems of contamination and equipment reliability are related to equipment design, construction and operation and, provided sufficiently rigorous standards are applied, these problems can be overcome. In fact, the fermentation industry has recognized the superiority of continuous culture for the production of biomass and several large-scale processes have been established. This aspect is considered in more detail in a later section of this Chapter.

METABOLITE PRODUCTIVITY

Theoretically, a fermentation to produce a metabolite should also be more productive in continuous culture than in batch because a continuous culture may be operated at the dilution rate which maintains product output at its maximum, whereas in batch culture product formation may be a transient phenomenon during the fermentation. The kinetics of product formation in continuous culture have been reviewed by Pirt (1975) and Trilli (1990). Product formation in a chemostat may be described as:

Change in product concentration = production − output:

or:

$$dp/dt = q_p x - Dp \tag{2.28}$$

where p is the concentration of product

and q_p is the specific rate of product formation (mg product g^{-1} biomass h^{-1}).

At steady state, $dp/dt = 0$, and thus:

$$\bar{p} = q_p \cdot x/D \tag{2.29}$$

where $\bar{p}$ is the steady-state product concentration. If q_p is strictlyrelated to μ, then as D increases so will q_p; thus, the steady-state product concentration ($\bar{p}$) and product output (Dp) will behave in the same way as x and Dx, as shown in Fig 2.10a. If q_p is independent of μ then it will be unaffected by D and thus concentration will decline with increasing D but output will remain constant, as shown in Fig. 2.10b. If product formation occurs only within a certain range of growth rates (dilution rates) then a more complex relationship is produced.

Thus, from this consideration a chemostat process for the production of a product can be designed to optimize either output (g dm^{-3} h^{-1}) or product concentration. However, as Heijnen *et al.* (1992) explained,

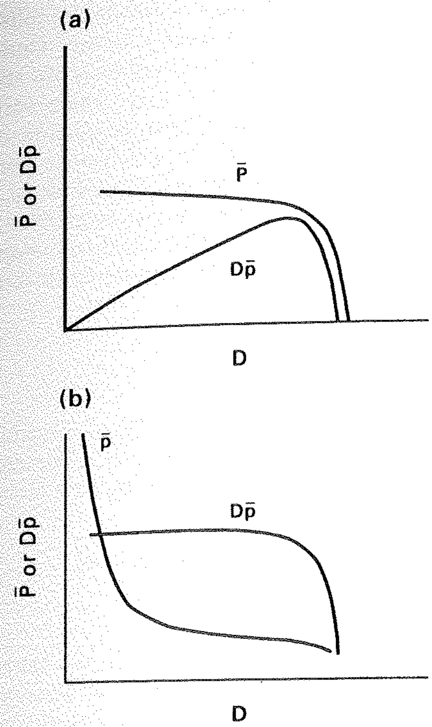

FIG. 2.10. The effect of D on steady-state product concentration ($\bar{p}$) and product output ($D_{\bar{p}}$) when:
(a) q_p is growth related.
(b) q_p is independent of growth rate.

when q_p is growth-related the advantage of high productivity obtained at high dilution rates must be balanced against the disadvantage of low product concentration resulting in increased downstream processing costs. The other arguments presented for the superiority of continuous culture for biomass production also hold true for product synthesis — ease of automation and the advantages of steady state conditions. The question that then arises is 'Why has the fermentation industry not adopted continuous culture for the manufacture of microbial products?' It can be appreciated that the arguments cited previously against continuous culture (contamination and equipment reliability) are not valid as these difficulties have been overcome in the large-scale continuous biomass processes. The answer to the question lies in the highly selective nature of continuous culture. We have already seen that μ is determined by D in a steady state chemostat and that μ and D are related to substrate concentration according to the equation:

$$D = \mu = \mu_{max}\bar{s}/(K_s + \bar{s})$$

The effect of substrate concentration on specific growth rate for two organisms, A and B, is shown in Fig. 2.11. A is capable of growing at a higher specific growth rate at any substrate concentration. The self-balancing properties of the chemostat mean that the organism reduces the substrate concentration to the value where $\mu = D$. Thus, at dilution rate X, organism A would reduce the substrate concentration to Z. However, at this substrate concentration, organism B could grow only at a μ of Y. Therefore, if organisms A and B were introduced into a chemostat operating at dilution rate X, A would reduce the substrate concentration to Z at which B could not maintain a μ of X and would be washed out at a rate of $(X - Y)$ and a monoculture of A would be established eventually. The same situation would occur if A and B were mutant strains arising from the same organism. Commercial organisms have been selected and mutated to produce metabolites at very high concentrations (see Chapter 3) and, as a result, tend to grow inefficiently with low μ_{max} values and, possibly, high K_s values. Back mutants of production strains produce much lower concentrations of product and, thus, grow more efficiently. If such back mutants arise in a chemostat industrial process, then the production strain will be displaced from the fermentation as described in the foregoing scenario.

Calcott (1981) described this phenomenon as "contamination from within" and this type of 'contamination' cannot be solved by the design of more 'secure' fermenters. Thus, it is the problem of strain degeneration which has limited the application of large scale continuous culture to biomass and, to a lesser extent, potable and industrial alcohol. The production of alcohol by continuous culture is feasible because it is a byproduct of energy generation and, thus, is not a drain on the resources of the organism. However, it is possi-

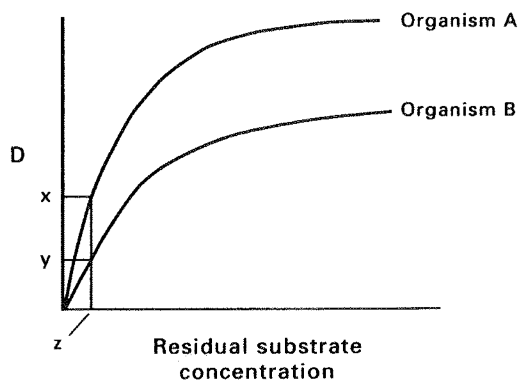

FIG. 2.11. Competition between two organisms in a chemostat.

ble that the technique could be exploited for other products provided strain degeneration is controlled; this may be possible in certain genetically engineered strains. It has been reported that the development of a continuous process for the manufacture of polyhydroxybutyrate, a biopolymer. Other processes have been developed using chemostat culture.

The adoption of continuous culture for animal cell products is even more complex than for microbial systems. Griffiths (1992) compared the following process options for producing an animal cell product:

(i) Batch culture.
(ii) Semi-continuous culture where a portion of the culture is harvested at regular intervals and replaced by an equal volume of medium.
(iii) Fed-batch culture where medium is fed to the culture resulting in an increase in volume (see later section)
(iv) Continuous perfusion where an immobilized cell population is perfused with fresh medium and is equivalent to an internal feedback continuous system.
(v) Continuous culture.

The characteristics of all five modes of operation are shown in Table 2.3, from which it may be seen that the perfusion continuous system appears extremely attractive. However, the practicalities of running a large scale continuous perfusion system present considerable difficulties. The process has to be reliable and able to operate aseptically for the long periods necessary to exploit the advantage of a continuous process. Also, the licensing of a continuous process may present some difficulties where a consignment of product must be traceable to a batch of raw materials. In a long-term continuous process several different batches of media would have to be used which presents the problem of associating product with raw material. Furthermore, it is difficult to monitor the genetic stability of cells which are immobilized in a large reactor system. Thus, large scale (kg quantities) animal cell products are still produced by batch methods. However, where very high value products are required and production can be satisfied on a small scale, the continuous perfusion system is a very attractive proposition (Griffiths, 1992).

Continuous brewing and biomass production which are the major industrial applications of continuous microbial culture will now be considered in more detail.

CONTINUOUS BREWING

The brewing industry in the United Kingdom has had a relatively brief 'courtship' with continuous culture. Two types of continuous brewing have been used:

(i) The cascade or multistage system.
(ii) The tower system.

Hough *et al.* (1976) described the cascade system utilized at Watneys' Mortlake brewery in London. The process utilized three vessels, the first two for fermentation and the third for separation of the yeast biomass. The specific gravity of the wort was reduced from 1040 to 1019 in the first vessel and from 1019 to 1011 in the second vessel. The residence time for the system was 15 to 20 hours, using worts in the specific gravity range of 1035 to 1040, and it could be run continuously for 3 months. However, it is believed that the system was abandoned due to problems of excessive biomass production. This process appears to have been used widely in New Zealand with greater success (Kirsop, 1982), but apparently newer installations are of the batch type.

A typical tower fermenter for brewing is illustrated in Fig. 2.12. The system is partially closed in that

TABLE 2.3. *Comparison of the performance of different operational modes of an animal cell fermentation* (*After* Griffiths, 1992)

Operational mode	Cell No. ($\times 10^{-6}$ cm^{-3})	Product yield (mg day^{-1})	Product yield (mg month^{-1})	Length of run (days)
Batch	3	100	200	7
Semi-continuous	3	200	600	21
Fed-batch	6	200	500	14
Perfusion	30 +	3000	12000	> 30
Continuous culture	2	300	1200	> 100

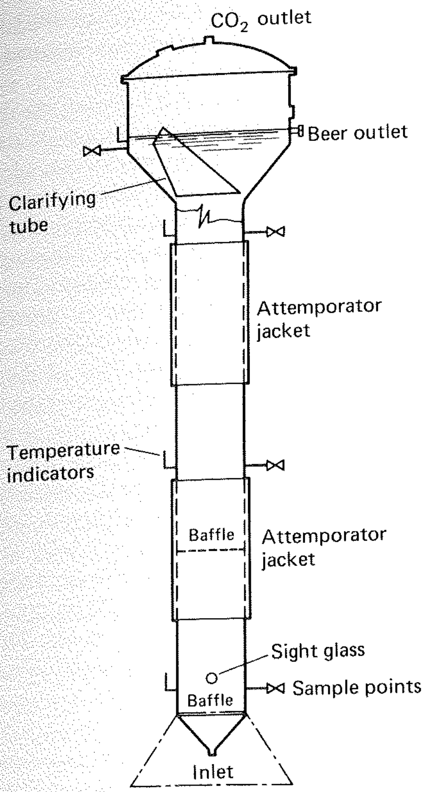

FIG. 2.12. A schematic representation of a tower fermenter for the brewing of beer (Royston, 1966).

relatively little yeast leaves the fermenter due to the highly flocculent nature of the strains employed. Thus, the system is a type of internal feed-back. Wort is introduced into the base of the tower and passes through a porous plug of yeast. As the wort rises through the vessel it is progressively fermented and leaves the fermenter via a yeast-separation zone, which is twice the diameter of the rest of the tower. Hough *et al.* (1976) described the protocol employed in the establishment of a yeast plug in the tower and the subsequent operating conditions. Prior to the fermentation, the tower is thoroughly cleaned and steam sterilized, aseptic operation being more important for the continuous process than the batch one. The vessel is filled partially with sterile wort and inoculated with a laboratory culture. The initial stages are designed to encourage high biomass production by the periodical addition of wort over about a 9-day period. A porous plug of yeast develops at the base of the tower. The flow rate of the wort is then gradually increased over a further 9 to 12 days, by which time an approximate steady state may be achieved. Following the establishment of a high biomass in the fermenter the system is operated such that the wort is converted to beer with the formation of approximately the same amount of yeast as would be produced in the batch process. The beer produced during the 3-week start-up period is usually below specification and would have to be blended with high-quality beer. Thus, more than 3 months' continuous operation is necessary to compensate for the initial losses of the process.

The major advantage of the continuous tower process was that the wort residence time could be reduced from about 1 week to 4 to 8 hours as compared with the batch system. However, the development of the cylindro-conical vessel (initially described by Nathan in the 1930s, but not introduced until the 1970s, see also Chapter 7) led to the shortening of the batch fermentation time to approximately 48 hours. Although this is still considerably longer than the residence time in a tower fermenter, it should be remembered that beer conditioning and packaging takes considerably longer than the fermentation stage, so that the difference in the overall processing time between the tower and the cylindro-conical batch process is not sufficient to justify the disadvantages of the tower. The major disadvantages of the tower system are the long start-up time, the technical complexity of the plant, the requirement for more highly skilled personnel than for a batch plant, the inflexibility of the system in that a long time delay would ensue between changing from one beer fermentation to another and, finally, the difficulty in matching the flavour of the continuously produced product with that of the traditional batch product. Thus, the continuous-tower system has fallen from use, with virtual universal adoption of the cylindro-conical batch process.

CONTINUOUS CULTURE AND BIOMASS PRODUCTION

Microbial biomass which is produced for human or animal consumption is referred to as single cell protein (SCP). Although yeast was produced as food on a large scale in Germany during the First World War (Laskin, 1977) the concept of utilizing microbial biomass as food was not thoroughly investigated until the 1960s. Since the 1960s, a large number of industrial companies have explored the potential of producing SCP from a wide range of carbon sources. Almost without exception, these investigations have been based on the use of continuous culture as the growth technique.

As previously discussed, continuous culture is the ideal method for the production of microbial biomass. The superior productivity of the technique, compared with that of batch culture, may be exploited fully and the problem of strain degeneration is not as significant as in the production of microbial metabolites. The selective pressure in the chemostat would tend to work to the advantage of the industrialist producing SCP, in that the most efficient strain of the organism would be selected, although this is not necessarily the case for mycelial processes. The development of SCP processes generated considerable research into large-scale chemostat design and the behaviour of the production organism in these very large vessels. Many 'novel' fermenters have been designed for SCP processes and these are considered in more detail in Chapter 7.

A very wide range of carbon sources have been investigated for the production of SCP. Whey has been used as a carbon source for biomass production since the 1940s and such fermentations have been shown to be economic in that they provide a high-grade feed product, whilst removing an, otherwise, troublesome waste product of the cheese industry (Meyrath and Bayer, 1979). Cellulose has been investigated extensively as a potential carbon source for SCP production and this work has been reviewed by Callihan *et al.* (1979) and Woodward (1987). The major difficulty associated with the use of cellulose as a substrate is its recalcitrant nature.

An enormous amount of research has been conducted into the use of hydrocarbons as sources of carbon for biomass processes; the hydrocarbons investigated being methane, methanol and *n*-alkanes. A large number of commercial firms were involved in this research field but very few created viable, commercial processes based on SCP production from hydrocarbons because of the economic difficulties involved (Sharp, 1989). At the start of this research, hydrocarbons were relatively cheap but, following the 1973 Middle East War, oil prices escalated and transformed the economic basis of biomass production from petroleum sources. ICI were successful in developing a commercial process for production of bacterial biomass (Pruteen) from methanol at an annual rate of 54,000 to 70,000 tonnes. The process utilized a novel air-lift, pressure cycle fermenter, of 3000 m^3 capacity, and was the first commercial process to produce SCP from methanol (King, 1982). The fermentation was run successfully for periods in excess of 100 days without contamination (Howells, 1982). Regrettably, the economics of the process were such that when the price of soya and fishmeal declined Pruteen could not compete as an animal feed. Selling in bulk ceased in 1985 (Sharp, 1989) and the 3000 m^3 vessel was eventually demolished.

The expertise developed by ICI during the Pruteen project and RHM's research into the use of a fungus, *Fusarium graminearum*, for the production of human food formed the basis of a joint venture between the two companies. The ICI pressure cycle pilot-plant was used to produce the fungal biomass (Myco-protein, marketed as Quorn) in continuous culture. The advantage of fungal biomass is that it may be processed to give a textured protein which is acceptable for human consumption. The low shear properties of the air-lift vessel conserve the desirable morphology of the fungus. The process is operated at a dilution rate of between 0.17 and 0.20 h^{-1} (μ_{max} is 0.28 h^{-1}). The phenomenon of mutation and intense selection in the chemostat has proved to be problematical in Myco-protein fermentation, because highly branched mutants have arisen in the vessel resulting in the loss of the desirable morphology. However, the process may still be operated in chemostat culture for 1000 hours on the full scale (Trinci, 1992).

Comparison of batch and continuous culture as investigative tools

Although the use of continuous culture on an industrial scale is very limited it is an invaluable investigative technique. The principle characteristic of batch culture is change. Even during the log phase cultural conditions are not constant and it is only the constant maximum specific growth rate which gives the semblance of stability — biomass concentration, substrate concentration and microbial products all change exponentially. During the deceleration phase the onset of nutrient limitation causes the growth rate to decline from its maximum to zero in a very short time, so it is virtually impossible to study the physiological effects of nutrient limitation in batch culture. As Trilli (1990) pointed out, adaptation of an organism to change is not instantaneous, so that the activity of a batch culture is not in equilibrium with the composition of its environment. Physiological events in a batch culture may have been initiated by a change in the environment which took place some significant time before the change was observed. Thus, it is very difficult to relate 'cause and effect'. The major feature of continuous culture, on the other hand, is 'the steady state' — biomass, substrate and product concentration should remain constant over very long periods of time. Specific growth rate is controlled by dilution rate and growth is nutrient limited. However, it is important not to exaggerate the significance of the steady state because a constant biomass level does not necessarily indicate that the culture is physiologically stable (Malek *et al.*, 1988). It is possible to separate the effects of growth rate and other environmental conditions, for example temperature, pH and dissolved oxygen concentration. Furthermore, because any of a wide range of substrates may be used to limit growth in the chemostat the effects of μ and substrate concentration may be distinguished.

Continuous cultures may generate valuable physiological information on an industrial strain which may be used in the optimization of the commercial process. An excellent example is the effect of growth rate and limiting substrate on metabolite formation. The interpretation of secondary metabolites as compounds produced in the idiophase of batch culture may lead one to suppose that the specific production rates (q_p) of such compounds are inversely linked to specific growth rate (μ). The testing of this supposition may be achieved in chemostat culture and it has been shown to be correct for cephamycin and thienamycin synthesis by *Streptomyces cattleya* (Lilley *et al.*, 1981) and gibberellin by *Gibberella fujikuroi* (Bu'Lock *et al.*, 1974). However, different relationships have been demonstrated in other systems. Pirt and Righelato (1967) and Ryu and Hospodka (1980) showed that the q_p of penicillin is positively correlated with μ up to a specific growth rate of 0.013 h^{-1}, after which it is independent of μ. Pirt (1990) suggests that the apparent negative correlation may be related to penicillin degradation. These observations suggest that the growth rate in a commercial penicillin process should not decline below 0.013 h^{-1}. Positive correlations between μ and q_p have been obtained for chlortetracycline production by *Streptomyces aureofaciens* (Sikyta *et al.*, 1961), oxytetracycline by *Streptomyces rimosus* (Rhodes, 1984) and erythromycin A by *Streptomyces erythraeus* (Trilli *et al.*, 1987).

The fiercely selective nature of the chemostat, which is its major disadvantage for industrial production, makes it an excellent tool for the isolation and improvement of micro-organisms. The use of continuous culture in this context is considered in Chapter 3, from which it may be seen that continuous enrichment culture offers considerable advantages over batch enrichment techniques and that continuous culture may be used very successfully to select strains producing higher yields of certain microbial enzymes.

FED-BATCH CULTURE

Yoshida *et al.* (1973) introduced the term fed-batch culture to describe batch cultures which are fed continuously, or sequentially, with medium, without the removal of culture fluid. A fed-batch culture is established initially in batch mode and is then fed according to one of the following feed strategies:

(i) The same medium used to establish the batch culture is added, resulting in an increase in volume.
(ii) A solution of the limiting substrate at the same concentration as that in the initial medium is added, resulting in an increase in volume.
(iii) A concentrated solution of the limiting substrate is added at a rate less than in (i) and (ii), resulting in an increase in volume.
(iv) A very concentrated solution of the limiting substrate is added at a rate less than in (i), (ii) and (iii), resulting in an insignificant increase in volume.

Fed-batch systems employing strategies (i) and (ii) are described as variable volume, whereas a system employing strategy (iv) is described as fixed volume. The use of strategy (iii) gives a culture intermediate between the two extremes of variable and fixed volume.

The kinetics of the two basic types of fed-batch culture, variable volume and fixed volume, will now be described.

Variable volume fed-batch culture

The kinetics of variable volume fed-batch culture have been developed by Dunn and Mor (1975) and Pirt (1974, 1975, 1979). The following account is based on that of Pirt (1975). Consider a batch culture in which growth is limited by the concentration of one substrate; the biomass at any point in time will be described by the equation:

$$x_t = x_0 + Y(S_R - s) \tag{2.30}$$

where x_t is the biomass concentration after time, t hours,
and x_0 is the inoculum concentration.

The final biomass concentration produced when $s = 0$ may be described as x_{max} and, provided that x_0 is small compared with x_{max}:

$$x_{max} \simeq Y \cdot S_R \tag{2.31}$$

If, at the time when $x = x_{max}$, a medium feed is started such that the dilution rate is less than μ_{max}, virtually all the substrate will be consumed as fast as it enters the culture, thus:

$$FS_R \simeq \mu(X/Y) \qquad (2.32)$$

where F is the flow rate of the medium feed,
and X is the total biomass in the culture, described by $X = xV$, where V is the volume of the culture medium in the vessel at time t.

From equation (2.32) it may be concluded that input of substrate is equalled by consumption of substrate by the cells. Thus, $(ds/dt) \simeq 0$. Although the total biomass in the culture (X) increases with time, cell concentration (x) remains virtually constant, that is $(dx/dt) \simeq 0$ and therefore $\mu \simeq D$. This situation is termed a quasi steady state. As time progresses the dilution rate will decrease as the volume increases and D will be given the expression:

$$D = F/(V_0 + Ft) \qquad (2.33)$$

where V_0 is the original volume. Thus, according to Monod kinetics, residual substrate should decrease as D decreases resulting in an increase in the cell concentration. However, over most of the range of μ which will operate in fed-batch culture, S_R will be much larger than K_s so that, for all practical purposes, the change in residual substrate concentration would be extremely small and may be considered as zero. Thus, provided that D is less than μ_{max} and K_s is much smaller than S_R, a quasi steady state may be achieved. The quasi steady state is illustrated in Fig. 2.13a. The major difference between the steady state of a chemostat and the quasi steady state of a fed-batch culture is that μ is constant in the chemostat but decreases in the fed-batch.

Pirt (1979) has expressed the change in product concentration in variable volume fed-batch culture in the same way as for continuous culture (see equation 2.28):

$$dp/dt = q_p x - Dp.$$

Thus, product concentration changes according to the balance between production rate and dilution by the feed. However, in the genuine steady state of a chemostat, dilution rate and growth rate are constant whereas in a fed-batch quasi steady state they change over the time of the fermentation. Product concentration in the chemostat will reach a steady state, but in a fed-batch system the profile of the product concentration over the time of the fermentation will be dependent on the relationship between q_p and μ (hence D). If q_p is strictly growth related then it will change as μ changes with D and, thus, the product concentration will remain constant. However, if q_p is constant and independent of μ, then product concentration will decrease at the start of the cycle when Dp is greater than $q_p x$ but will rise with time as D decreases and $q_p x$ becomes greater than Dp. These relationships are shown in Fig. 2.14a. If q_p is related to μ in a complex manner, then the product concentration will vary according to that relationship. Thus, the feed strategy of a fed-batch system would be optimized according to the relationship between q_p and μ.

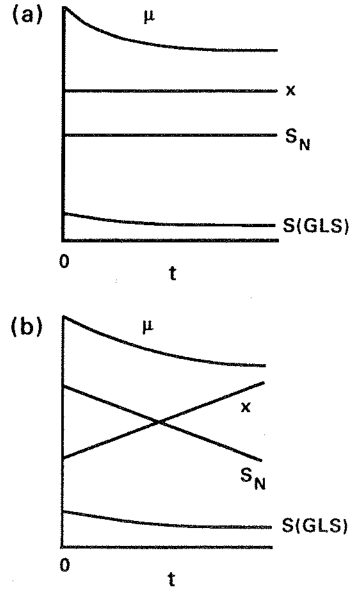

FIG. 2.13. Time profiles of fed-batch cultures.
μ = specific growth rate
x = biomass concentration
S(GLS) = growth limiting substrate
S_N = any other substrate than S(GLS)
(a) Variable volume fed-batch culture.
(b) Fixed volume fed-batch culture.
(Pirt, 1979).

Fixed volume fed-batch culture

Pirt (1979) described the kinetics of fixed volume fed-batch culture as follows. Consider a batch culture in which the growth of the process organism has depleted the limiting substrate to a limiting level. If the limiting substrate is then added in a concentrated feed such that the broth volume remains almost constant, then:

$$dx/dt = GY \qquad (2.34)$$

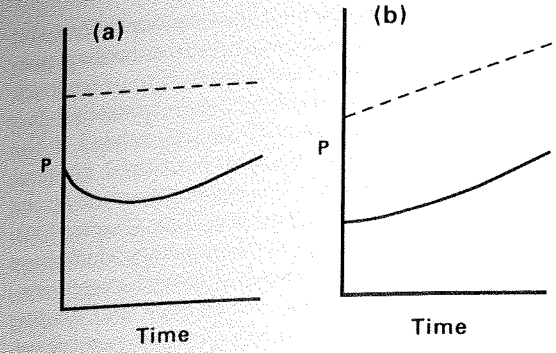

FIG. 2.14. Product concentration (p) in fed-batch culture when q_p is growth related (– – – – –) or non-growth related, i.e. q_p constant (———)
(a) Variable volume fed-batch culture.
(b) Fixed volume fed-batch culture.
(Modified from Pirt, 1979.)

where G is the substrate feed rate (g dm^{-3} h^{-1})
and Y is the yield factor.
But $dx/dt = \mu x$, thus substituting for dx/dt in equation (2.34) gives:

$$\mu x = GY, \text{ and thus:}$$

$$\mu = GY/x \qquad (2.35)$$

Provided that GY/x does not exceed μ_{max} then the limiting substrate will be consumed as soon as it enters the fermenter and $ds/dt \simeq 0$. However, dx/dt may not be equated to zero, as in the case of variable volume fed-batch, because the biomass concentration, as well as the total amount of biomass in the fermenter, will increase with time. Biomass concentration is given by the equation:

$$x_t = x_{\text{å}} + GYt \qquad (2.36)$$

where x_t is the biomass after operating in fed batch for t hours
and $x_{\text{å}}$ is the biomass concentration at the onset of fed-batch culture.

As biomass increases then the specific growth rate will decline according to equation (2.35). The behaviour of a fixed volume fed-batch culture is illustrated in Fig. 2.13b from which it may be seen that μ declines (according to equation (2.35)), the limiting substrate concentration remains virtually constant, biomass increases and the concentration of the non-limiting nutrients declines.

Pirt (1979) described the product balance in a fixed volume fed-batch system as:

$$dp/dt = q_p x,$$

but substituting for x from equation (2.36) gives:

$$dp/dt = q_p(x_{\text{å}} + GYt).$$

If q_p is strictly growth-rate related then product concentration will rise linearly as for biomass. However, if q_p is constant then the rate of increase in product concentration will rise as growth rate declines, i.e. as time progresses and x increases. These relationships are shown in Fig. 2.14b. If q_p is related to μ in a complex manner then the product concentration will vary according to that relationship. As in the case of variable volume fed-batch the feed profile would be optimized according to the relationship between q_p and μ.

Cyclic fed-batch culture

The life of a variable volume fed-batch fermentation may be extended beyond the time it takes to fill the fermenter by withdrawing a portion of the culture and using the residual culture as the starting point for a further fed-batch process. The decrease in volume results in a significant increase in the dilution rate (assuming that the flow rate remains constant) and thus, eventually, in an increase in the specific growth rate. The increase in μ is then followed by its gradual decrease as the quasi steady state is re-established. Such a cycle may be repeated several times resulting in a series of fed-batch fermentations. Thus, the organism would experience a periodic shift-up in growth rate followed by a gradual shift-down. This periodicity in growth rate may be achieved in fixed volume fed-batch systems by diluting the culture when the biomass reaches a concentration which cannot be maintained under aerobic conditions. Dilution results in a decline in x and, thus, according to equation (2.35) an increase in μ. Subsequently, as feeding continues, the growth rate will decline gradually as biomass increases and approaches the maximum sustainable in the vessel once more, at which point the culture may be diluted again. Dilution would be achieved by withdrawing culture and refilling to the original level with sterile water or medium not containing the feed substrate.

Application of fed-batch culture

The use of fed-batch culture by the fermentation industry takes advantage of the fact that the concentration of the limiting substrate may be maintained at a very low level, thus avoiding the repressive effects of high substrate concentration. Furthermore, the fed-batch system also gives some control over the organisms' growth rate, which is also related to the specific oxygen uptake rate giving some control over the oxygen demand of the fermentation (see Chapter 9). Both variable and fixed volume systems result in low limiting substrate concentrations, but the quasi steady-state of the variable volume system has the advantage of maintaining the concentrations of both the biomass and the non-limiting nutrients constant. Pirt (1979) cites this as an important feature because the concentrations of substrates other than those which limit growth can have a significant effect on biomass composition and product formation.

The obvious advantage of cyclic fed-batch culture is that the productive phase of a process may be extended under controlled conditions. However, a further advantage lies in the controlled periodic shifts in growth rate which may provide an opportunity to optimize product synthesis. Dunn and Mor (1975) pointed out that changes in the rates of chemical processes can give rise to increases in intermediate concentrations and similar effects may be possible in microbial systems. This observation is particularly relevant to secondary metabolite production which is maximal in batch culture during the deceleration phase. Bushell (1989) suggested that optimum conditions for secondary metabolite synthesis may occur during the transition phase after the withdrawal of a volume of broth from the vessel and before the re-establishment of the steady-state following the resumption of the nutrient feed. During this period the dilution rate will be greater than the growth rate but, according to Bushell, the rate of uptake of the growth-limiting substrate should respond immediately to the increased substrate concentration. Thus, an imbalance results between the substrate uptake rate and the specific growth rate — this imbalance then contributing to a diversion of intermediates into secondary metabolism.

The advantages of cyclic fed-batch culture must be weighed against difficulties inherent in the system. Care has to be taken in the design of cyclic fed-batch processes to ensure that toxins do not accumulate to inhibitory levels and that nutrients other than those incorporated into the feed medium become limiting (Queener and Swartz, 1979). Also, a prolonged series of fed-batch cycles may result in the accumulation of non-producing or low-producing variants.

The early fed-batch systems that were developed did not incorporate any form of feedback control and relied on the inherent quasi steady state to maintain process stability. However, the use of concentrated feeds resulting in very high biomass concentrations and the development of more sophisticated feeding programmes has necessitated the introduction of feedback control techniques. In such feedback controlled fermentations a process parameter directly related to the organism's physiological state is monitored continuously by an on-line sensor. The signal generated by the sensor is then used in a control loop (see Chapter 8) to control the medium feed rate. Parameters which have been utilized in this way include dissolved oxygen concentration, pH, effluent gas composition and limiting substrate concentration. Examples of these control systems are included in the next section and in Chapter 9.

Examples of the use of fed-batch culture

Fed-batch culture was used in the production of bakers' yeast as early as 1915. It was recognized that an excess of malt in the medium would lead to too high a growth rate resulting in an oxygen demand in excess of that which could be met by the equipment. This resulted in the development of anaerobic conditions and the formation of ethanol at the expense of biomass production (Reed and Nagodawithana, 1991). Thus, the organism was grown in an initially weak medium to which additional medium was added at a rate less than the maximum rate at which the organism could use it. Thus, this process fulfils the criteria stipulated in Pirt's (1975) kinetic description for the establishment of a quasi steady state, that is, a substrate limited culture and the use of a feed rate equivalent to a dilution rate less than μ_{max}. It is now recognized that bakers' yeast is very sensitive to free glucose and respiratory activity may be repressed at a concentration of about 5 μg dm^{-3} (Crabtree, 1929). Thus, a high glucose concentration represses respiratory activity as well as giving rise to a high growth rate, the oxygen demand of which cannot be met. In modern fed-batch processes for yeast production the feed of molasses is under strict control based on the automatic measurement of traces of ethanol in the exhaust gas of the fermenter. Although such systems may result in low growth rates, the biomass yield is near the theoretical maximum obtainable (Fiechter, 1982).

It is interesting to note that the production of recombinant proteins from yeast may be achieved using fed-batch culture techniques very similar to that developed for the bakers' yeast fermentation. Gu *et al.* (1991) reported the production of hepatitis B surface antigen (HBsAg) in a 0.9 dm^3 fed-batch reactor where the feed rate was increased exponentially to maintain a relatively high growth rate. HBsAg production was growth associated in this strain and good productivity was achieved by maintaining a high growth rate whilst keeping the glucose concentration below that which would repress respiratory activity. Ibba *et al.* (1993) reported a cyclic fed-batch process for recombinant hirudin from *S. cerevisiae*, under the control of a constitutive promoter. The cyclic fed-batch process gave three times the hirudin activity of a continuous fermentation, the superior productivity being due to increased transcription although a genetic explanation of the results could not be offered.

Penicillin fermentation provides an excellent example of the use of feed systems in the production of a secondary metabolite. The fermentation may be divided into two phases — the 'rapid-growth' phase, during which the culture grows at μ_{max}, and the 'slow-growth' or 'production' phase. Glucose feeds may be used to control the metabolism of the organism during both phases. During the rapid-growth phase an excess of glucose causes an accumulation of acid and a biomass oxygen demand greater than the aeration capacity of the fermenter, whereas glucose starvation may result in the organic nitrogen in the medium being used as a carbon source, resulting in a high pH and inadequate biomass formation (Queener and Swartz, 1979). The accumulation of hexose may be prevented by the use of a slowly hydrolysed carbohydrate such as lactose in simple batch culture (Matelova, 1976). However, according to Queener and Swartz, considerable increases in productivity have been achieved by the use of computer controlled feeding of glucose during the rapid growth phase, such that the dissolved oxygen or pH is maintained within certain limits. Both control parameters essentially measure the same activity in that both oxygen concentration and pH will fall when glucose is in excess, due to an increased respiration rate and the accumulation of organic acids when the respiration rate exceeds the aeration capacity of the fermenter. Both systems appear to work well in controlling feed rates during the rapid-growth phase.

During the production phase of the penicillin fermentation the feed rates utilized should limit the growth rate and oxygen consumption such that a high rate of penicillin synthesis is achieved, and sufficient dissolved oxygen is available in the medium. The control factor in this phase is normally dissolved oxygen because pH is less responsive to the effect of dissolved oxygen on penicillin synthesis than on growth. As the fed-batch process proceeds then the total biomass, viscosity and oxygen demand increase until, eventually, the fermentation is oxygen limited. However, limitation may be delayed by reducing the feed rate as the fermentation progresses and this may be achieved by the use of computer controlled systems.

Suzuki *et al.* (1987) developed a pH feedback fed-batch system for the production of thiostrepton from *Streptomyces laurentii*. When glucose was exhausted in the fermentation the pH rose immediately and this event was used as the signal for the addition of more feed which consisted of a concentrated glucose, corn steep liquor, soy bean meal and mineral mixture. This process maintained a biomass level of 157 g dm^{-3} and a thiostrepton concentration of 10.5 g dm^{-3} with a productivity nine times that of a conventional batch culture.

Many enzymes are subject to catabolite repression, where enzyme synthesis is prevented by the presence of rapidly utilized carbon sources (Aunstrup *et al.*, 1979). It is obvious that this phenomenon must be avoided in enzyme fermentations and fed-batch culture is the major technique used to achieve this. Concentrated medium is fed to the culture such that the carbon source does not reach the threshold for catabolite repression. For example, Waki *et al.* (1982) controlled the production of cellulase by *Trichoderma reesei* in fed-batch culture utilizing CO_2 production as the control factor and Suzuki *et al.* (1988) achieved high lipase production from *Pseudomonas fluorescens* also using CO_2 production to control the addition of an oil feed.

Shioya (1990) developed a method for the optimization of fed-batch systems based on the relationship between μ and q_p, the product specific production rate. Once the relationship between the two parameters was established a computer control system was used to maintain the fed-batch at the optimum specific growth rate (feed rate). The system was tested for a number of fermentations including histidine (*Brevibacterium flavum*), acid phosphatase and glutathione (*S. cerevisiae*), and lysine (*Corynebacterium glutamicum*). Specific growth rate was maintained constant using a feed-forward control profile which was updated throughout the fermentation as data were collected. Very promising results were obtained but difficulties were experienced in generating sufficiently accurate data to up-date the control system and maintain the specific growth rate constant.

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CHAPTER 3

The Isolation, Preservation and Improvement of Industrially Important Micro-organisms

THE ISOLATION OF INDUSTRIALLY IMPORTANT MICRO-ORGANISMS

THE MOST publicized advances in biotechnology over the last ten years have been those in recombinant DNA technology. Whilst these advances have resulted in the development of extremely valuable new commercial processes and have improved many others, it is worthwhile reiterating Buckland's (1992) comment that the combined sales of four new microbial secondary metabolites introduced in the 1980s was greater than the sales of all the recombinant products added together. Thus, the diversity of micro-organisms may be exploited still by searching for strains from the natural environment able to produce products of commercial value. The first stage in the screening for micro-organisms of potential industrial application is their isolation. Isolation involves obtaining either pure or mixed cultures followed by their assessment to determine which carry out the desired reaction or produce the desired product. In some cases it is possible to design the isolation procedure in such a way that the growth of producers is encouraged or that they may be recognized at the isolation stage, whereas in other cases organisms must be isolated and producers recognized at a subsequent stage. However, it should be remembered that the isolate must eventually carry out the process economically and therefore the selection of the culture to be used is a compromise between the productivity of the organism and the economic constraints of the process. Bull *et al.* (1979) cited a number of criteria as being important in the choice of organism:

1. The nutritional characteristics of the organism. It is frequently required that a process be carried out using a very cheap medium or a pre-determined one, e.g. the use of methanol as an energy source. These requirements may be met by the suitable design of the isolation medium.
2. The optimum temperature of the organism. The use of an organism having an optimum temperature above 40° considerably reduces the cooling costs of a large-scale fermentation and, therefore, the use of such a temperature in the isolation procedure may be beneficial.
3. The reaction of the organism with the equipment to be employed and the suitability of the organism to the type of process to be used.
4. The stability of the organism and its amenability to genetic manipulation.
5. The productivity of the organism, measured in its ability to convert substrate into product and to give a high yield of product per unit time.
6. The ease of product recovery from the culture.

Points 3, 4, and 6 would have to be assessed in detailed tests subsequent to isolation and the organism most well suited to an economic process chosen on the basis of these results. However, before the process may be

TABLE 3.1. *Major culture collections*

Culture collection	Address
National Collection of Type Cultures (NCTC)	PHLS Central Public Health Laboratory, 61 Colindale Avenue, London NW9 5HT, UK
National Collections of Industrial and Marine Bacteria Ltd. (NCIB, NCMB)	23 St Machar Drive, Aberdeen AB2 1RY, UK
National Collection of Yeast Cultures (NCYC)	AFRC Institute of Food Research, Norwich Laboratory, Colney Lane, Norwich NR4 7UA, UK
Collection of International Mycological Institute (IMI)	Culture Collection and Industrial Service Division, Ferry Lane, Kew, Surrey TW9 3AF, UK
American Type Culture Collection (ATCC)	12301 Parklawn Drive, Rockville, MD 20852, USA
Deutsche Sammlung von Mikroorganismen und Zelkulturen (DSM)	Mascheroder Weg 1 b, D-3300 Braunschweig, Germany
Centraalbureau voor Schimmelcultures (CBS)	P.O. Box 273, Oosterstraat 1, NL-3740 AG Baarn, Netherlands
Czechoslovak Collection of Microorganisms (CCM)	Masaryk University, Jostova 10, 662 43 Brno, Czech Republic
Collection Nationale de Cultures de Microorganisms (CNCM)	Institut Pasteur, 25, rue du Docteur Roux, F-75724 Paris Cedex 15, France
Japan Collection of Microorganisms (JCM)	Riken, Wako-shi, Saitama, 351-01 Japan
Culture Collection of the Institute for Fermentation (IFO)	Institute for Fermentation, 17-85 Jugo-Honchmachi, 2-chome, Yodogawa-ku, Osaka, Japan

put into commercial operation the toxicity of the product and the organism has to be assessed.

The above account implies that cultures must be isolated, in some way, from natural environments. However, the industrial microbiologist may also 'isolate' micro-organisms from culture collections. Kirsop and Doyle (1991) have provided a comprehensive list of collections and Table 3.1 cites some examples. Such collections may provide organisms of known characteristics but may not contain those possessing the most desirable features, whereas the environment contains a myriad of organisms, very few of which may be satisfactory. It is certainly cheaper to buy a culture than to isolate from nature, but it is also true that a superior organism may be found after an exhaustive search of a range of natural environments. The economic considerations are discussed in more detail in Chapter 12. However, it is always worthwhile to purchase cultures demonstrating the desired characteristics, however weakly, as they may be used as model systems to develop culture and assay techniques which may then be applied to the assessment of natural isolates.

The ideal isolation procedure commences with an environmental source (frequently soil) which is highly probable to be rich in the desired types, is so designed as to favour the growth of those organisms possessing the industrially important characteristic (i.e. the industrially useful characteristic is used as a selective factor) and incorporates a simple test to distinguish the most desirable types. Selective pressure may be used in the isolation of organisms which will grow on particular substrates, in the presence of certain compounds or under cultural conditions adverse to other types. However, if it is not possible to apply selective pressure for the desired character it may be possible to design a procedure to select for a microbial taxon which is known to show the characteristic at a relatively high frequency, e.g. the production of antibiotics by streptomycetes. Alternatively, the isolation procedure may be designed to exclude certain microbial 'weeds' and to encourage the growth of more novel types. Indeed, as pointed out by Bull (1992) for screening programmes to continue to generate new products it is becoming increasingly more important to concentrate on lesser known microbial taxa or to utilize very specific screening tests to identify the desired activity. During the 1980s significant advances have been made in the establishment of taxonomic databases describing the properties of microbial groups and these databases have been used to predict the cultural conditions which would select for the growth of particular taxa. Thus, the advances in the taxonomic description of taxa have allowed the rational design of procedures for the isolation of strains which may have a high probability of being productive or are representatives of unusual groups. The advances in pharmacology and molecular biology have also enabled the design of more effective screening tests to identify productive strains amongst the isolated organisms.

Isolation methods utilizing selection of the desired characteristic

In this section we consider methods which take direct advantage of the industrially relevant property exhibited by an organism to isolate that organism from an environmental source. Isolation methods depending on the use of desirable characteristics as selective factors are essentially types of enrichment culture. Enrichment culture is a technique resulting in an increase in the number of a given organism relative to the numbers of other types in the original inoculum. The process involves taking a mixed population and providing conditions either suitable for the growth of the desired type, or unsuitable for the growth of the other organisms, e.g. by the provision of particular substrates or the inclusion of certain inhibitors. Prior to the culture stage it is often advantageous to subject the environmental source (normally soil) to conditions which favour the survival of the organisms in question. For example, air-drying the soil will favour the survival of actinomycetes.

ENRICHMENT LIQUID CULTURE

Enrichment liquid culture is frequently carried out in shake flasks. However, the growth of the desired type from a mixed inoculum will result in the modification of the medium and therefore changes the selective force which may allow the growth of other organisms, still viable from the initial inoculum, resulting in a succession. The selective force may be re-established by inoculating the enriched culture into identical fresh medium. Such sub-culturing may be repeated several times before the dominant organism is isolated by spreading a small inoculum of the enriched culture onto solid medium. The time of sub-culture in an enrichment process is critical and should correspond to the point at which the desired organism is dominant.

The prevalence of an organism in a batch enrichment culture will depend on its maximum specific growth rate compared with the maximum specific growth rates of the other organisms capable of growth in the inoculum. Thus, provided that the enrichment broth is sub-cultured at the correct times, the dominant organism will be the fastest growing of those capable of growth. However, it is not necessarily true that the organism with the highest specific growth rate is the most useful, for it may be desirable to isolate the organism with the highest affinity for the limiting substrate.

The problems of time of transfer and selection on the basis of maximum specific growth rate may be overcome by the use of a continuous process where fresh medium is added to the culture at a constant rate. Under such conditions the selective force is maintained at a constant level and the dominant organism will be selected on the basis of its affinity for the limiting substrate rather than its maximum growth rate.

The basic principles of continuous culture are considered in Chapter 2 from which it may be seen that the growth rate in continuous culture is controlled by the dilution rate and is related to the limiting substrate concentration by the equation:

$$\mu = \mu_{\max} s/(K_s + s). \qquad (3.1)$$

Equation (3.1) is represented graphically in Fig. 3.1. A model of the competition between two organisms capable of growth in a continuous enrichment culture is represented in Fig. 3.2. Consider the behaviour of the two organisms, A and B, in Fig. 3.2. In continuous culture the specific growth rate is determined by the substrate concentration and is equal to the dilution rate, so that at dilution rates below point Y in Fig. 3.2 strain B would be able to maintain a higher growth rate than strain A, whereas at dilution rates above Y strain A would be able to maintain a higher growth rate. Thus, if A and B were present in a continuous enrichment culture, limited by the substrate depicted in Fig. 3.2, strain A would be selected at dilution rates above Y and strain B would be selected at dilution rates below Y. Thus, the organisms which are isolated by continuous enrichment culture will depend on the dilution rate employed which may result in the isolation of

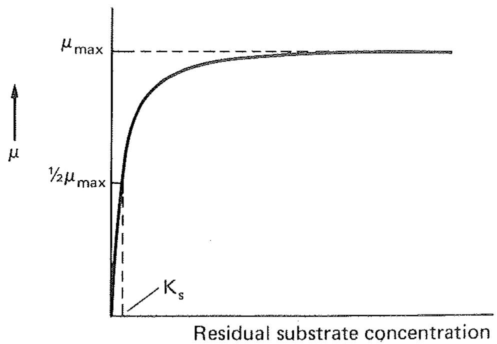

FIG. 3.1. The effect of substrate concentration on the specific growth rate of a micro-organism.

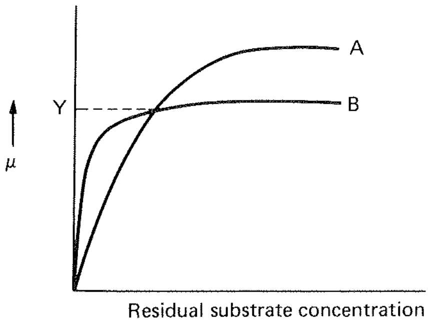

FIG. 3.2. The effect of substrate concentration on the specific growth rates of two micro-organisms A and B.

organisms not so readily recovered by batch techniques.

Continuous enrichment techniques are especially valuable in isolating organisms to be used in a continuous-flow commercial process. Organisms isolated by batch enrichment and purification on solid media frequently perform poorly in continuous culture (Harrison *et al.*, 1976), whereas continuous enrichment provides an organism, or mixture of organisms, adapted to continuous culture. The enrichment procedure should be designed such that the predicted isolate meets as many of the criteria of the proposed process as possible and both Johnson (1972) and Harrison (1978) have discussed such procedures for the isolation of organisms to be used for biomass production. Johnson emphasized the importance of using the carbon source to be employed in the subsequent commercial process as the sole source of organic carbon in the enrichment medium, and that the medium should be carbon limited. The inclusion of other organic carbon sources, such as vitamins or yeast extract, may result in the isolation of strains adapted to using these, rather than the principal carbon source, as energy sources. The isolation of an organism capable of growth on a simple medium should also form the basis of a cheaper commercial process and should be more resistant to contamination — a major consideration in the design of a commercial continuous process. The use of as high as possible an isolation temperature should also result in the isolation of a strain presenting minimal cooling problems in the subsequent process.

The main difficulty in using a continuous-enrichment process is the washout of the inoculum before an adapted culture is established. Johnson (1972) suggested that the isolation process should be started in batch culture using a 20% inoculum and as soon as growth is observed, the culture should be transferred to fresh medium and the subsequent purification and stabilization of the enrichment performed in continuous culture. The continuous system should be periodically inoculated with soil or sewage which may not only be a source of potential isolates but should also ensure that the dominant flora is extremely resistant to contamination.

Harrison (1978) proposed two solutions to the problem of early washout in continuous isolation processes: The first uses a turbidostat and the second uses a two-stage chemostat (see Chapter 2). A turbidostat is a continuous-flow system provided with a photoelectric cell to determine the turbidity of the culture and maintain the turbidity between set points by initiating or terminating the addition of medium. Thus, washout is avoided as the medium supply will be switched off if the biomass falls below the lower fixed point. The use of a turbidostat will result in selection on the basis of maximum specific growth rate as it operates at high levels of limiting substrate. Thus, although the use of the turbidostat removes the danger of washout it is not as flexible a system as the chemostat which may be used at a range of dilution rates. The two-stage chemostat described by Harrison (1978) is very similar to Johnson's (1972) procedure. The first stage of the system was used as a continuous inoculum for the second stage and consisted of a large bottle containing a basic medium inoculated with a soil infusion. Continuous inoculation was employed until an increasing absorbance was observed in the second stage. Bull (1992) advocated the use of feed-back continuous systems for the isolation of strains with particularly high affinity for substrate and this approach would also guard against premature washout.

The use of continuous enrichment culture has frequently resulted in the selection of stable, mixed cultures presumably based on some form of symbiotic relationship. It is extremely unlikely that such mixed, stable systems could be isolated by batch techniques so that the adoption of continuous enrichment may result in the development of novel, mixed culture fermentations. Harrison *et al.* (1972) isolated a mixed culture using methane as the carbon source in a continuous enrichment and demonstrated that the mixture contained one methylotroph and a number of non-methylotrophic symbionts. The performance of the methylotroph in pure culture was invariably poorer than the mixture in terms of growth rate, yield and culture stability.

Continuous enrichment has also been used for the isolation of organisms to be used in systems other than biomass production; Rowley and Bull (1977) used the technique to isolate an *Arthrobacter* sp. producing a yeast lysing enzyme complex. The technique has been used widely for the

ENRICHMENT CULTURES USING SOLIDIFIED MEDIA

Solidified media have been used for the isolation of certain enzyme producers and these techniques usually involve the use of a selective medium incorporating the substrate of the enzyme which encourages the growth of the producing types. Aunstrup *et al.* (1972) isolated species of *Bacillus* producing alkaline proteases. Soils of various pHs were used as the initial inoculum and, to a certain extent, the number of producers isolated correlated with the alkalinity of the soil sample. The soil samples were pasteurized to eliminate non-sporulating organisms and then spread onto the surface of agar media at pH 9–10, containing a dispersion of an insoluble protein. Colonies which produced a clear zone due to the digestion of the insoluble protein were taken to be alkaline protease producers. The size of the clearing zone could not be used quantitatively to select high producers as there was not an absolute correlation between the size of the clearing zone and the production of alkaline protease in submerged culture. However, this example demonstrates the importance of choice of starting material, the use of a selective force in the isolation and the incorporation of a preliminary diagnostic test, albeit of limited use.

Isolation methods not utilizing selection of the desired characteristic

The synthesis of some products does not give the producing organism any selective advantage which may be exploited directly in an isolation procedure. Examples include the production of antibiotics and growth promoters. Therefore, a pool of organisms has to be isolated and subsequently tested for the desired characteristic. The major problem faced by industrial microbiologists in this situation is the reisolation of strains which have already been screened many times before. However, this 'reinvention of the wheel' syndrome may be minimized in two major ways:

(i) Developing procedures to favour the isolation of unusual taxa which are less likely to have been screened previously.

(ii) Identifying selectable features correlated with the unselectable industrial trait thus enabling an enrichment process to be developed.

Much use has been made of numerical taxonomic data bases to design media selective for particular taxa. For example, Williams' group at Liverpool University (U.K.) used such databases to design isolation media to either encourage the growth of uncommon streptomycetes or to discourage the growth of *Streptomyces albidoflavus* (Vickers *et al.*, 1984; Williams and Vickers, 1988). Several groups of workers have taken advantage of the antibiotic sensitivity information stored in taxonomic databases to design media selective for particular taxa, as shown in Table 3.2. The incorporation of particular antibiotics into isolation media may result in the selection of the resistant taxa. Such techniques are reviewed by Goodfellow and O'Donnell (1989) and Bull (1992). Bull emphasized that the taxonomic approach may be optimized and developed according to Fig. 3.1. The isolates from an isolation procedure would be screened for activity and then the growth requirements of the positive cultures determined. This knowledge can then be used to optimize the isolation medium and the cycle begins again.

Whilst taxonomic databases are a convenient source of information, it is important to appreciate Huck *et al.*'s (1991) observation that these systems were designed to provide information for taxonomic differentiation within a group. Thus, some of the diagnostic data may not be applicable to isolation systems. More significantly, the reactions of organisms outside the taxon in question would not be listed. Of course, an environmental sample contains a vast variety of organisms and the design of isolation media based on knowledge of only one taxon may inadvertently result in the preferential isolation of undesirable types. Huck *et al.* (1991) used the statistical stepwise discrimination analysis (SDA) technique to design media for the positive selection of antibiotic producing soil isolates. This was achieved by characterizing a collection of eubacterial and actinomycete soil isolates according to 43 physiological and nutritional tests. Certain features were identified which, when used as selective factors, enhanced the probability of either isolating actinomycetes or antibiotic producing actinomycetes. These features are shown in Table 3.3. Using this approach several media were developed which enhanced the isolation of antibiotic producers.

The most desirable isolation medium would be one

TABLE 3.2 *Antibacterial compounds used in selective media for the isolation of actinomycetes* (Goodfellow and O'Donnell, 1989)

Selective agent	Target organism	Reference
Bruneomycin	*Micromonospora*	Preobrazhenskayai *et al.* (1975)
Dihydroxymethylfuratriazone	*Microtetraspora*	Tomita *et al.* (1980)
Gentamycine	*Micromonospora*	Bibikova *et al.* (1981)
Kanamycin	*Actinomadura*	Chormonova (1978)
Nitrofurazone	*Streptomyces*	Yoshioka (1952)
Novobiocin	*Micromonospora*	Sveshnikova *et al.* (1976)
Tellurite	*Actinoplanes*	Willoughby (1971)
Tunicamycin	*Micromonospora*	Wakkisaka *et al.* (1982)

which selects for the desired types and also allows maximum genetic expression. Cultures grown on such media could then be used directly in a screen. However, it is more common that, once isolated, the organisms are grown on a range of media designed to enhance productivity. Nisbet (1982) put forward some guidelines for the design of such media and these are summarized in Table 3.4.

Screening methods

Early screening strategies tended to be empirical, labour intensive and had relatively low success rates. As the number of commercially important compounds isolated increased, the success rates of such screens decreased further. Thus, new screening methods have been developed which are more precisely targeted to identify the desired activity. The evolution of antibiotic screens serves as an excellent illustration of the development of more precise, targeted systems.

Antibiotics were initially detected by growing the potential producer on an agar plate in the presence of an organism (or organisms) against which antimicrobial action was required. Production of the antibiotic was detected by inhibition of the test organism(s). Alternatively, the microbial isolate could be grown in liquid culture and the cell-free broth tested for activity. This approach was extended by using a range of organisms to detect antibiotics with a defined antibacterial spectrum. For example, Zahner (1978) discussed the use of test organisms to detect the production of antibiotics with confined action spectra. The use of *Bacillus subtilis* and *Streptomyces viridochromogenes* or *Clostridium pasteurianum* allows the identification of antibiotics with a low activity against *B. subtilis* and a high activity against the other test organisms. Such antibiotics may be new because they would not be isolated by the more common tests using *B. subtilis* alone. The kirromycin group of antibiotics were discovered using methods of this type.

In the 15 years prior to 1971, no novel naturally occurring β-lactams were discovered (Nisbet and Porter, 1989). However, the advances made in the

TABLE 3.3 *Selective substrates for the isolation of actinomycetes and antibiotic-producing actinomycetes*

Substrates selective for Actinomycetes	Substrates selective for antibiotic-producting actinomycetes
Proline	Proline
Glucose (1.0%)	Glucose (1.0%)
Glycerol	Glycerol
Starch	Starch
Humic acid (0.1%)	Humic acid (0.1%)
Propionate (0.1%)	Zinc
Methanol	Alanine
Nitrate	Potassium
Calcium	Vitamins
	Cobalt (0.05%)
	Phenol (0.01%)
	Asparagine

TABLE 3.4 *Guidelines for 'overproduction media'* (Nisbet, 1982)

1. Prepare a range of media in which different types of nutrients become growth-limiting e.g. C, N, P, O
2. For each type of nutrient depletion use different forms of the growth-sufficient nutrient
3. Use a polymeric or complexed form of the growth-limiting nutrient
4. Avoid the use of readily assimilated forms of carbon (glucose) or nitrogen (NH_4^+) that may cause catabolite repression
5. Ensure that known cofactors are present (Co^{3+}, Mg^{2+}, Mn^{2+}, Fe^{2+}).
6. Buffer to minimize pH changes.

understanding of cell-wall biosynthesis and the mode of action of antibiotics allowed the development of mode of action screens in the 1970s which resulted in a very significant increase in the discoveries of new β-lactam antibiotics. Nagarajan *et al.*'s (1971) discovery of the cephamycins was based on the detection of compounds which induced morphological changes in susceptible bacteria. The appreciation that the mode of action of penicillins was the inhibition of the transpeptidase enzyme crosslinking mucopeptide molecules led to the development of enzyme inhibitor assays, which were particularly attractive because they could be automated. Fleming *et al.* (1982) described the devlopment of an automated screen for the detection of carboxypeptidase inhibitors which has led to the detection of several novel cephamycin and carbapenem compounds.

The increasing frequency of penicillin and cephalosporin resistance amongst clinical bacteria led to the development of mechanism based screens for the isolation of more effective antibacterials. The logic of the Beechams Pharmaceuticals group of Brown *et al.* (1976) was to search for a compound which would inhibit β-lactamase and could be incorporated with ampicillin as a combination therapeutic agent. Samples were tested for their ability to increase the inhibitory effect of ampicillin on a β-lactamase producing *Klebsiella aerogenes* and this strategy resulted in the discovery of clavulanic acid. Further examples of targeted antibiotic screens are given in the excellent review by Nisbet and Porter (1989).

The concept of using the inhibition of enzymes as a screening mechanism was pioneered by Umezawa (1972) in his search for microbial products inhibiting key enzymes of human metabolism. His approach was based on the logic that, if a compound inhibits a key human enzyme *in vitro*, it may have a pharmacological action *in vivo*. Such screens have been applied to a wide range of pharmacological targets and have resulted in the isolation of several important drugs. From *Aspergillus*, Alberts *et al.* (1980) isolated mevinolin which is an inhibitor of hydroxy-methyl-glutaryl reductase, the rate limiting step in the biosynthesis of sterols. It is now marketed as an agent to lower high cholesterol levels. Bull (1992) list the following clinical situations in which microbial products have been shown to inhibit key enzymes: hypercholesterolaemia, hypertension, gastric inflammation, muscular dystrophy, benign prostate hyperplasia and systemic lupus erythemosus. A further specific example is provided by the work of Hashimoto *et al.* (1990). The activity of carbapenem antibiotics is lost in therapy due to renal dehydropeptidase activity. These workers isolated microbial products capable of inhibiting the enzyme which could then be administered along with the antibiotic to maintain its clinical activity.

The detection of pharmacological agents by receptor–ligand binding assays has been developed rapidly by pharmaceutical companies (Bull, 1992). These are extensions of the enzyme inhibitor approach but agents which block receptor sites are likely to be more effective at very low concentrations. The gastrointestinal hormone cholecystokinin (CCK) controls a range of digestive activities such as pancreatic secretion and gall bladder contraction. Receptor screening identified a fungal metabolite, aperlicin (from *Aspergillus alliaceus*) that had a very high affinity for CCK receptors. Although the fungal product did not prove to be a suitable drug it was used as a model for the design of analogues which were receptor binders and pharmacologically acceptable.

The progress in molecular biology, genetics and immunology has also contributed extensively to the development of innovative screens, by enabling the construction of specific detector strains, increasing the availability of enzymes and receptors and constructing extremely sensitive assays. Bull (1992) summarized the major contributions as follows:

(i) The provision of test organisms that have increased sensitivities, or resistances, to known agents. For example, the use of super-sensitive strains for the detection of β-lactam antibiotics.

(ii) The cloning of genes coding for enzymes or receptors that may be used in inhibitor or binding screens makes such materials more accessible and available in much larger amounts.

(iii) The development of reporter gene assays. A reporter gene is one which codes for an easily assayable product so that it can be used to detect the activation of a control sequence to which it is fused. Such systems have been used in the search for metabolites that disrupt viral replication.

(iv) Molecular probes for particular gene sequences may enable the detection of organisms capable of producing certain product groups. This information may be used to focus the search on these organisms in an attempt to find novel representatives of an already known commercially attractive chemical family.

(v) The development of immunologically based assays such as ELISA.

It is important to appreciate at this stage that the advances in the biological sciences which have enabled the design of sophisticated screening tests have been paralleled by the development of robotic automation systems. This means that the engineering now exists to automate such tests, resulting in an enormous throughput. Indeed, Nisbet and Porter (1989) claimed that "the modern microbial discovery programme must achieve rates of 10^5 tests per year, for as many as 20 different assay systems, to compete in the race for novel therapeutics."

THE PRESERVATION OF INDUSTRIALLY IMPORTANT MICRO-ORGANISMS

The isolation of a suitable organism for a commercial process may be a long and very expensive procedure and it is therefore essential that it retains the desirable characteristics that led to its selection. Also, the culture used to initiate an industrial fermentation must be viable and free from contamination. Thus, industrial cultures must be stored in such way as to eliminate genetic change, protect against contamination and retain viability. An organism may be kept viable by repeated sub-culture into fresh medium, but, at each cell division, there is a small probability of mutations occurring and because repeated sub-culture involves very many such divisions, there is a high probability that strain degeneration would occur. Also, repeated sub-culture carries with it the risk of contamination. Thus, preservation techniques have been developed to maintain cultures in a state of 'suspended animation' by storing either at reduced temperature or in a dehydrated form. Full details of the techniques are given by Kirsop and Doyle (1991).

Storage at reduced temperature

STORAGE ON AGAR SLOPES

Cultures grown on agar slopes may be stored in a refrigerator (5°) or a freezer (−20°) and sub-cultured at approximately 6-monthly intervals. The time of sub-culture may be extended to I year if the slopes are covered with sterile medicinal grade mineral oil.

STORAGE UNDER LIQUID NITROGEN

The metabolic activities of micro-organisms may be reduced considerably by storage at the very low temperatures (−150° to −196°) which may be achieved using a liquid nitrogen refrigerator. Snell (1991) claimed that this aproach is the most universally applicable of all preservation methods. Fungi, bacteriophage, viruses, algae, yeasts, animal and plant cells and tissue cultures have all been successfully preserved. The technique involves growing a culture to the maximum stationary phase, resuspending the cells in a cryoprotective agent (such as 10% glycerol) and freezing the suspension in sealed ampoules before storage under liquid nitrogen. Some loss of viability is suffered during the freezing and thawing stages but there is virtually no loss during the storage period. Thus, viability may be predictable even after a period of many years. Snell (1991) suggested that liquid nitrogen is the method of choice for the preservation of valuable stock cultures and may be the only suitable method for the long term preservation of cells that do not survive freeze-drying. Although the equipment is expensive the process is economical on labour. However, the method has the major disadvantage that liquid nitrogen evaporates and must be replenished regularly. If this is not done, or the apparatus fails, then the consequences are the loss of the collection.

Storage in a dehydrated form

DRIED CULTURES

Dried soil cultures have been used widely for culture preservation, particularly for sporulating mycelial organisms. Moist, sterile soil may be inoculated with a culture and incubated for several days for some growth to occur and then allowed to dry at room temperature for approximately 2 weeks. The dry soil may be stored in a dry atmosphere or, preferably, in a refrigerator. The technique has been used extensively for the storage of fungi and actinomycetes and Pridham *et al.* (1973) observed that of 1800 actinomycetes dried on soil about 50% were viable after 20-years storage.

Malik (1991) described methods which extend the approach using substrates other than soil. Silica gel and porcelain beads are suggested alternatives and detailed methods are given for these simple, inexpensive techniques in Malik's discussion.

LYOPHILIZATION

Lyophilization, or freeze-drying, involves the freezing of a culture followed by its drying under vacuum, which results in the sublimation of the cell water. The technique involves growing the culture to the maximum stationary phase and resuspending the cells in a protective medium such as milk, serum or sodium glutamate. A few drops of the suspension are transferred to an ampoule, which is then frozen and subjected to a high vacuum until sublimation is complete, after which the ampoule is sealed. The ampoules may be stored in a refrigerator and the cells may remain viable for 10 years or more (Perlman and Kikuchi, 1977).

Lyophilization is very convenient for service culture collections (Snell, 1991) because, once dried, the cultures need no further attention and the storage equipment (a refrigerator) is cheap and reliable. Also, the freeze dried ampoules may be dispatched as such, still in a state of 'suspended animation' whereas liquid nitrogen stored cultures begin to deteriorate. However, freeze-dried cultures are tedious to open and revitalize and several sub-cultures may be needed before the cells regain their typical characteristics. Overall, the technique appears to be second only to liquid nitrogen storage and even when liquid nitrogen is used makes an excellent insurance against the possibility of the breakdown of the nitrogen freezer. The technique is considered in detail by Rudge (1991).

Quality control of preserved stock cultures

Whichever technique is used for the preservation of an industrial culture it is essential to be certain of the quality of the stocks. Each batch of newly preserved cultures should be routinely checked to ensure their quality and such a procedure has been outlined by Lincoln (1960): a single colony of the culture to be preserved is inoculated into a shake flask and the growth of the culture observed to ensure a typical growth pattern; after a further shake-flask sub-culture the broth is used to prepare a large number of storage ampoules. At least 3% of the ampoules are reconstituted and the cultures assessed for purity, viability and productivity. If the samples fail any one of these tests the entire batch should be destroyed. Thus, by the use of such a quality-control system stock cultures may be retained, and used, with confidence.

THE IMPROVEMENT OF INDUSTRIAL MICRO-ORGANISMS

Natural isolates usually produce commercially important products in very low concentrations and therefore every attempt is made to increase the productivity of the chosen organism. Increased yields may be achieved by optimizing the culture medium and growth conditions, but this approach will be limited by the organism's maximum ability to synthesize the product. The potential productivity of the organism is controlled by its genome and, therefore, the genome must be modified to increase the potential yield. The cultural requirements of the modified organism would then be examined to provide conditions that would fully exploit the increased potential of the culture, while further attempts are made to beneficially change the genome of the already improved strain. Thus, the process of strain improvement involves the continual genetic modification of the culture, followed by reappraisals of its cultural requirements.

Genetic modification may be achieved by selecting natural variants, by selecting induced mutants and by selecting recombinants. There is a small probability of a genetic change occurring each time a cell divides and when it is considered that a microbial culture will undergo a vast number of such divisions it is not surprising that the culture will become more heterogeneous. The heterogeneity of some cultures can present serious problems of yield degeneration because the variants are usually inferior producers compared with the original culture. However, variants have been isolated which are superior producers and this has been observed frequently in the early stages in the development of a natural product from a newly isolated organism. An explanation of this phenomenon for mycelial organisms may be that most new isolates are probably heterokaryons (contain more than one type of nucleus) and the selection of the progeny of uninucleate spores results in the production of homokaryons (contain only one type of nucleus) which may be superior producers. However, the phenomenon is also observed with unicellular isolates which are certainly not heterokaryons. Therefore, it is worthwhile to periodically plate out the producing culture and screen a proportion of the progeny for productivity; this practice has the added advantage that the operator tends to become familiar with morphological characteristics associated with high productivity and, by selecting 'typical' colonies, a strain subject to yield degeneration may still be used with consistent results.

Therefore, selection of natural variants may result in increased yields but it is not possible to rely on such improvements, and techniques must be employed to increase the chances of improving the culture. These techniques are the isolation of induced mutants and recombination. The most dramatic examples of strain improvement come from the applications of recombinant DNA technology which has resulted in organisms producing compounds which they were not able to produce previously. Furthermore, the advances in these techniques have resulted in very significant improvements in the production of conventional fermentation products. However, it should be remembered that these methods have not replaced mutant isolation but have made an invaluable addition to an impressive repertoire. The techniques of mutant isolation have contributed enormously to the development of present-day industrial strains and these techniques will be considered first. The applications of recombination systems will then be discussed along with the interactions between mutant isolation and recombination in strain improvement advances.

Dulaney and Dulaney (1967) compared the spread in productivity of chlortetracycline of natural variants of *Streptomyces viridifaciens* with the spread in productivity of the survivors of an ultraviolet treatment. The results of their comparison are shown in Figs 3.3 and 3.4, from which it may be seen that although there are more inferior producers amongst the survivors of the ultraviolet treatment there are also strains producing more than twice the parental level, far greater than the best of the natural variants. The use of ultraviolet light is only one of a large number of physical or chemical agents which increase the mutation-rate — such agents are termed mutagens. The reader is referred to Baltz (1986) and Birge (1988) for accounts of the modes of action of mutagens. The vast majority of induced muta-

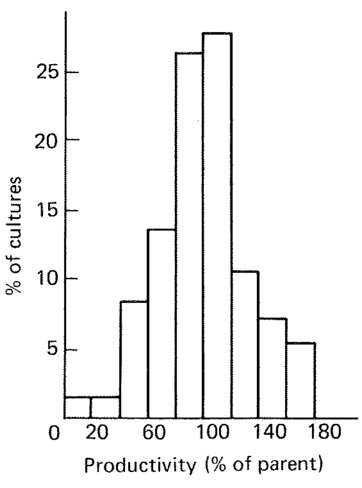

FIG. 3.3. The spread in productivity of chortetracycline of natural variants of *Streptomyces viridifaciens* (Dulaney and Dulaney, 1967).

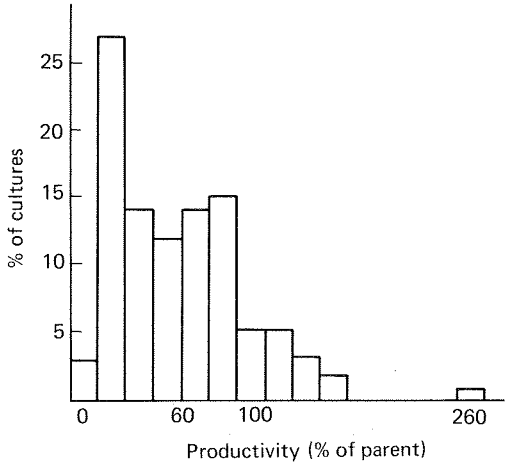

FIG. 3.4. The spread in chlortetracycline productivity of the survivors of a UV-treated population of *Streptomyces viridifaciens* (Dulaney and Dulaney, 1967).

tions are deleterious to the yield of the desired product but, as shown in Fig. 3.3, a minority are more productive than the parent. The problem of obtaining the high-yielding mutants may be approached from two theoretical standpoints; the number of desirable mutants may be increased by 'directed mutation', i.e. the use of a technique which will preferentially produce particular mutants at a high rate; or techniques may be developed to improve the separation of the few desirable types from the large number of mediocre producers.

Inherent in the concept of directed mutation is the assumption that a mutation programme can be optimized to produce mutants of a particular kind. The choice of mutagen was demonstrated to affect the success of mutation programmes early in the history of strain improvement schemes. For example, Hostalek (1964) claimed that ultraviolet radiation was the most effective mutagen for increasing the yield of tetracycline by strains of *Streptomyces aureofaciens*. DeWitt *et al.* (1989) emphasized that as well as certain mutagens being more beneficial, the dose will affect the generation of the desired types. Despite these observations it is frequently the case that it is difficult to predict what type of mutation is required at the molecular level to improve a strain, and therefore it is extremely unlikely that the concept of directed mutation can be applied in these circumstances. Thus, it is the second approach specified above that is likely to provide the solution to this type of problem, i.e. the development of selection techniques.

However, the concept of directed mutation is of considerable relevance when the genes to be modified are known and the organism is genetically well documented (Shortle *et al.*, 1981). In these systems a cloned DNA molecule may be subjected to *in vitro* enzymatic cleavage and manipulation such that the mutation is present at a particular site. Thus, reintroduction of the modified DNA into the cell should result in the production of a cell mutated at a particular site. The incorporation of a selectable marker into the cloned DNA would assist in the recovery of cells incorporating the modified gene(s). Thus, such systems still depend on the development of methods to recover cells displaying the desirable feature.

The separation of desirable mutants from the other survivors of a mutation treatment is similar to the isolation of desirable organisms from nature. Where possible, the mutant isolation procedure should use the improved characteristic of the desired mutant as a selective factor. Presumably, superior productivity is a result of a diversion of precursors into the product and/or a modification of the control mechanisms limiting the level of production. Thus, a knowledge of the biosynthetic route and the mechanisms of control of the biosynthesis of the product should enable the prediction of a 'blueprint' of the desirable mutant. Such a 'blueprint' might then enable the design of isolation techniques which would give the desired mutants a selective advantage over the other types present. Knowledge of biosynthetic routes and control mechanisms are more detailed for primary metabolites and, therefore, the use of selective pressure in mutant isolation is more common in the fields of amino acid, nucleotide and enzyme production than in secondary metabolite production. However, considerable progress has also been made in the design of such procedures for the isolation of secondary metabolite over-producers.

The selection of induced mutants synthesizing improved levels of primary metabolites

Before considering the methods used for the selection of mutants producing improved levels of primary metabolites it is necessary to study the mechanisms of control of their biosynthesis such that the 'blueprints', referred to above, may be drawn accurately. The levels of primary metabolites in micro-organisms are regulated by feedback control systems. The major systems involved are feedback inhibition and feedback repression. Feedback inhibition is the situation where the end product of a biochemical pathway inhibits the activity of an enzyme catalysing one of the reactions (normally the first reaction) of the pathway. Inhibition acts by the end product binding to the enzyme at an allosteric site which results in interference with the attachment of the enzyme to its substrate. Feedback repression is the situation where the end product (or a derivative of the end product) of a biochemical pathway prevents the synthesis of an enzyme (or enzymes) catalysing a reaction (or reactions) of the pathway. Repression occurs at the gene level by a derivative of the end product combining with the genome in such a way as to prevent the transcription of the gene into messenger RNA, thus resulting in the prevention of enzyme synthesis.

Feedback inhibition and repression frequently act in concert in the control of biosynthetic pathways, where inhibition may be visualized as a rapid control which switches off the biosynthesis of an end product and repression as a mechanism to then switch off the synthesis of temporarily redundant enzymes. The control of pathways giving rise to only one product (i.e. unbranched pathways) is normally achieved by the first enzyme in the sequence being susceptible to inhibition by the end product and the synthesis of all the enzymes being susceptible to repression by the end product, as shown in Fig. 3.5.

The control of biosynthetic pathways giving rise to a number of end products (branched pathways) is more complex than the control of simple, unbranched sequences. The end products of the same, branched biosynthetic pathway are rarely required by the micro-organism to the same extent, so that if an end product exerts control over a part of the pathway common to two, or more, end products then the organism may suffer deprivation of the products not participating in the control. Thus, mechanisms have evolved which enable the level of end products of branched pathways to be controlled without depriving the cell of essential intermediates. The following descriptions of these mechanisms are based on the effect of the control, which may be arrived at by inhibition, repression or a combination of both systems.

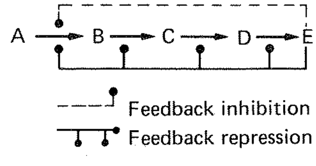

FIG. 3.5. The control of a biosynthetic pathway converting precursor A to end product E via the intermediates B, C and D.

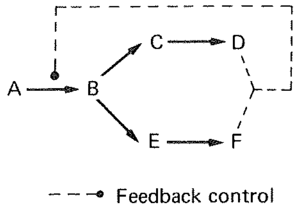

FIG. 3.6. The control of a biosynthetic pathway by the concerted effects of products D and F on the first enzyme of the pathway.

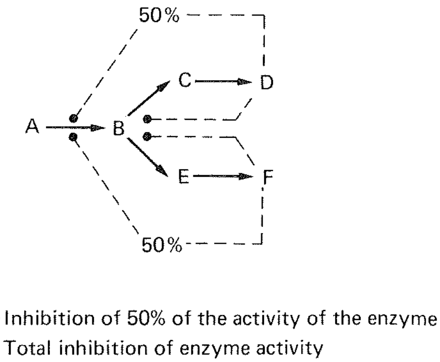

FIG. 3.8. The control of a biosynthetic pathway by the cumulative control of products D and F.

Concerted or multivalent feedback control. This control system involves the control of the pathway by more than one end product — the first enzyme of the pathway is inhibited or repressed only when all end products are in excess, as shown in Fig. 3.6.

Co-operative feedback control. The system is similar to concerted control except that weak control may be effected by each end product independently. Thus, the presence of all end products in excess results in a synergistic repression or inhibition. The system is illustrated in Fig. 3.7 and it may be seen that for efficient control to occur when one product is in excess there should be a further control operational immediately after the branch point to the excess product. Thus, the reduced flow of intermediates will be diverted to the product which is still required.

Cumulative feedback control. Each of the end products of the pathway inhibits the first enzyme by a certain percentage independently of the other end products. In Fig. 3.8 both D and F independently reduce the activity of the first enzyme by 50%, resulting in total inhibition when both products are in excess. As in the case of co-operative control, each end product must exert control immediately after the branch point so that the common intermediate, B, is diverted away from the pathway of the product in excess.

Sequential feedback control. Each end product of the pathway controls the enzyme immediately after the branch point to the product. The intermediates which then build up as a result of this control earlier enzymes in the pathway. Thus, in Fig. 3.9, D inhibits the conversion of B to C, and F inhibits the conversion of B to E. The inhibitory action of D, F, or both, would result in an accumulation of B which, in turn, would inhibit the conversion of A to B.

Isoenzyme control. Isoenzymes are enzymes which catalyse the same reaction but differ in their control characteristics. Thus, if a critical control reaction of a pathway is catalysed by more than one isoenzyme, then the different isoenzymes may be controlled by the different end products. Such a control system should be very efficient, provided that control exists immediately after the branch point so that the reduced flow of intermediates is diverted away from the product in excess. An example of the system is shown in Fig. 3.10.

Thus, the levels of microbial metabolites may be controlled by a variety of mechanisms, such that end products are synthesized in amounts not greater than those required for growth. However, the ideal industrial micro-organism should produce amounts far greater than those required for growth and, as sug-

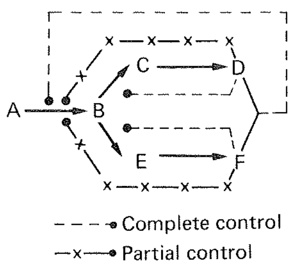

FIG. 3.7. The control of a biosynthetic pathway by the co-operative control by end products D and F.

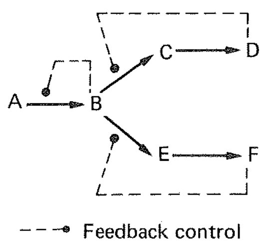

FIG. 3.9. The control of a biosynthetic pathway by sequential feedback control.

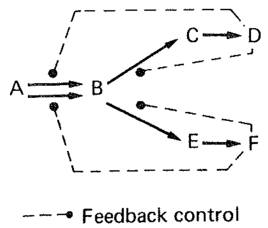

FIG. 3.10. The control of two isoenzymes (catalysing the conversion of A to B) by end products D and F.

gested earlier, an understanding of the control of production of a metabolite may enable the construction of a 'blueprint' of the most useful industrial mutant, i.e. one where the production of the metabolite is not restricted by the organism's control system. Such postulated mutants may be modified in three ways:

1. The organism may be modified such that the end products which control the key enzymes of the pathway are lost from the cell due to some abnormality in the permeability of the cell membrane.
2. The organism may be modified such that it does not produce the end products which control the key enzymes of the pathway.
3. The organism may be modified such that it does not recognize the presence of inhibiting or repressing levels of the normal control metabolites.

MODIFICATION OF THE PERMEABILITY

The best example of the modification of the permeability of a micro-organism is provided by the glutamic acid fermentation. Kinoshita *et al.* (1957a) isolated a biotin-requiring, glutamate-producing organism, subsequently named *Corynebacterium glutamicum*, the permeability of which could be modified by the level of biotin. Provided that the level of biotin in the production medium was below 5 μg dm^{-3} then the organism would excrete glutamate, but at concentrations of biotin optimum for growth the organism produced lactate (Kinoshita and Nakayama, 1978). Thus, the permeability of *C. glutamicum* may be controlled by the composition of the culture medium and, as such, is not an example of strain improvement in the sense that the term has been used in this chapter. However, the isolation of *C. glutamicum* was a major advance in the microbial production of amino acids and the demonstration of the role of permeability in the production of glutamate suggested the possibility of the genetic modification of permeability to achieve high levels of productivity. Thus, it is relevant at this stage to consider the physiology of glutamate production by biotin-limited culture.

Kinoshita's isolate was not only deficient in its ability to produce biotin but also in the level of α-ketoglutarate dehydrogenase, which normally converts α-ketoglutarate to succinate in the tricarboxylic acid cycle. Thus, a metabolic block results in the accumulation of a large concentration of α-ketoglutarate which the organism converts to glutamic acid. Oxaloacetate is regenerated in glutamate producers by the activity of the glyoxylate pathway as shown in Fig. 3.11.

When *C. glutamicum* was grown in a medium containing a high concentration of biotin, the organism synthesized glutamate at a level of 25–36 μg mg^{-1} dry weight of cells, further production being assumed to be prevented by some form of feedback control by glutamate of its own synthesis (Demain and Birnbaum,

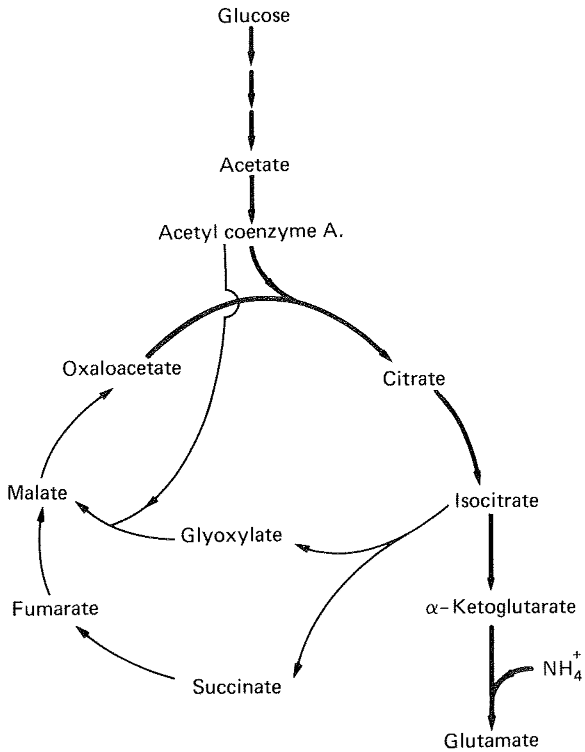

FIG. 3.11. Biosynthesis of glutamate by *C. glutamicum*. Heavy lines indicate the route to glutamate and the light lines indicate the route in the regeneration of oxaloacetate via the glyoxylate cycle.

1968). Under conditions of biotin limitation, glutamate was released from the cells and accumulated to a level of up to 50 g dm^{-3}. The increased permeability of the biotin-limited cells to glutamate corresponds with a change in the fatty acid and phospholipid content of the cell envelopes. The crucial factor in the control of permeability appears to be the synthesis of membranes deficient in phospholipids (Nakao *et al.*, 1973).

The role of biotin in the glutamate fermentation obviously necessitates the inclusion of limiting levels of biotin in the medium. Thus, the use of crude carbon sources such as cane molasses, which are rich in biotin, presented difficulties which have been overcome by the modification of the permeability by means other than biotin limitation. The inclusion of penicillin or fatty acid derivatives such as polyoxyethylene sorbitan mono-oleate (Tween 80) during logarithmic growth in biotin-rich medium caused an aberration of the cell envelope permeability resulting in the release of glutamate (Udagawa *et al.*, 1962).

The production of glutamate from hydrocarbons (as carbon sources) has also presented difficulties in the control of permeability, in that the assimilation of hydrocarbons results in the production of fatty acids which effectively bypasses the site of biotin control (Nakao *et al.*, 1972). The permeability of the producing organism may be controlled by the addition of penicillin (Wang *et al.*, 1979) but a genetic solution to this problem has also been found. Nakao *et al.* (1970) isolated a glycerol requiring auxotrophic mutant of *Corynebacterium alkanolyticum* in which phospholipid synthesis was controlled by the supply of glycerol. The mutant produced about 40 g dm^{-3} glutamate when grown on n-paraffins in the presence of 0.01% glycerol (Nakao *et al.*, 1972). Thus, an understanding of the mode of action of biotin limitation in the glutamate fermentation has led to the use of genetic modification of permeability as a method to overcome the normal control mechanisms of the producing organism.

THE ISOLATION OF MUTANTS WHICH DO NOT PRODUCE FEEDBACK INHIBITORS OR REPRESSORS

Mutants which do not produce certain feedback inhibitors or repressors may be useful for the production of intermediates of unbranched pathways and intermediates and end products of branched pathways. Demain (1972) presented several 'blue-prints' of hypothetical mutants producing intermediates and end products of biosynthetic pathways and these are illustrated in Fig. 3.12. The mutants illustrated in Fig. 3.12 do not produce some of the inhibitors or repressors of the pathways considered and, thus, the control of the pathway is lifted, but, because the control factors are also essential for growth, they must be incorporated into the medium at concentrations which will allow growth to proceed but will not evoke the normal control reactions.

In the case of Fig. 3.12(1) the unbranched pathway is normally controlled by feedback inhibition or repression of the first enzyme of the pathway by the end product, E. However, the organism represented in Fig. 3.12(1) is auxotrophic for E due to the inability to convert C to D so that control of the pathway is lifted and C will be accumulated provided that E is included in the medium at a level sufficient to maintain growth but insufficient to cause inhibition or repression.

Figure 3.12(2) is a branched pathway controlled by the concerted inhibition of the first enzyme in the pathway by the combined effects of E and G. The mutant illustrated is auxotrophic for E due to an inability to convert C to D, resulting in the removal of the concerted control of the first enzyme. Provided that E is included in the medium at a level sufficient to allow growth but insufficient to cause inhibition then C will be accumulated due to the control of the end product G on the conversion of C to F. The example shown in Fig. 3.12(3) is similar to that in Fig. 3.12(2) except that it is a double auxotroph and requires the feeding of both E and G. Figure 3.12(4) is, again, the same pathway and illustrates another double mutant with the deletion for the production of G occurring between F and G, resulting in the accumulation of F.

Figure 3.12(5) illustrates the accumulation of an end product of a branched pathway which is normally controlled by the feedback inhibition of the first enzyme in the pathway by the concerted effects of E and I. The mutant illustrated is auxotrophic for I and G due to an inability to convert C to F and, thus, provided G and I are supplied in quantities which will satisfy growth requirements without causing inhibition, the end product, E, will be accumulated.

All the hypothetical examples discussed above are auxotrophic mutants and, under certain circumstances, may accumulate relatively high concentrations of intermediates or end products. Therefore, the isolation of auxotrophic mutants may result in the isolation of high-producing strains, provided that the mutation for auxotrophy occurs at the correct site, e.g. between C and D in Figs 3.12(1) and (2). The recovery of auxotrophs is a simpler process than is the recovery of high producers, as such, so that the best approach is to

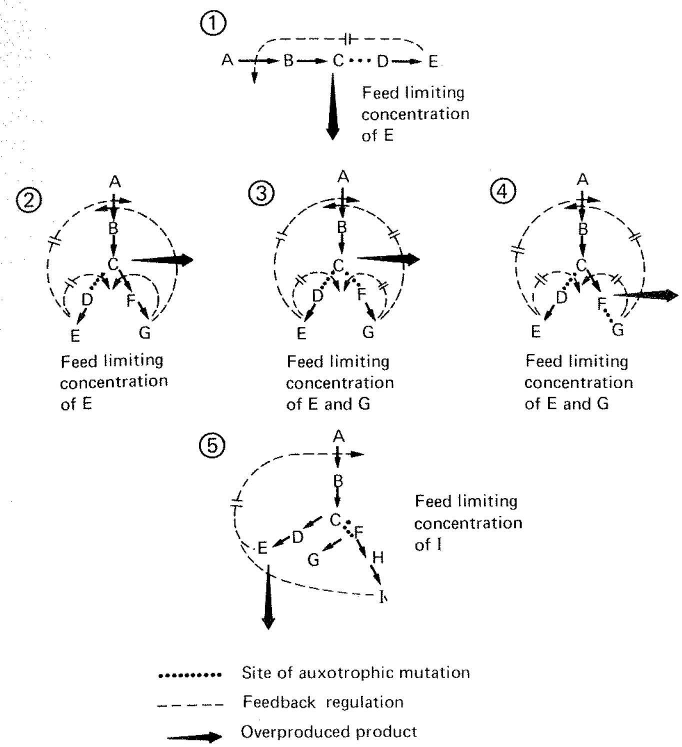

FIG. 3.12. Overproduction of primarymetabolites by decreasing the concentration of a repressing or inhibiting end product. ···· Site of auxotrophicmutations; ––––,feedback regulation; → overproduced product (Demain, 1972).

design a procedure to select relevant auxotrophs from the survivors of a mutation and subsequently screen the selected auxotrophs for productivity. Productive strains amongst the auxotrophs may be detected by over-layering colonies of the mutants with agar suspensions of bacteria auxotrophic for the required product. The high-producing mutants may be identified by the growth of the overlay around the producer. The most commonly used methods for the recovery of auxotrophic mutants are the use of some form of enrichment culture or the use of a technique to visually identify the mutants.

The enrichment processes employed are based on the provision of conditions which adversely affect the prototrophic cells but do not damage the auxotrophs. Such conditions may be achieved by exposing the population, in minimal medium, to an antimicrobial agent which only affects dividing cells which should result in the death of the growing prototrophs but the survival of the non-growing auxotrophs. Several techniques have been developed using different antimicrobials suitable for use with a range of micro-organisms.

Davis (1949) developed an enrichment technique utilizing penicillin as the inhibitory agent. The survivors of a mutation treatment were first cultured in complete medium, harvested by centrifugation, washed and resuspended in minimal medium plus penicillin. Only the growing prototrophic cells were susceptible to the penicillin and the non-growing auxotrophs survived. The cells were harvested by centrifugation, washed (to remove the penicillin and products released from lysed cells) and resuspended in complete medium to allow the growth of the auxotrophs, which could then be purified on solid medium. The nature of the auxotrophs isolated may be determined by the design of the so-called complete medium; if only one addition is made to the minimal medium then mutants auxotrophic for the additive should be isolated.

Abe (1972) described the use of Davis' technique to isolate auxotrophic mutants of the glutamic acid producing organism *C. glutamicum*. The procedure is outlined in Fig. 3.13.

Advantage has also been taken of the fact that the ungerminated spores of some organisms are more resistant to certain compounds than are the germinated spores. Thus, by culturing mutated spores in minimal

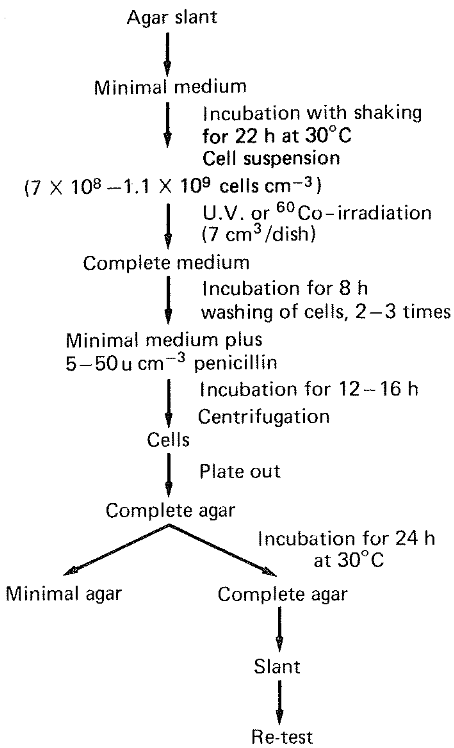

FIG. 3.13. The use of the penicillin selection method for the isolation of auxotrophic mutants of *C. glutamicum* (Abe, 1972).

medium only the prototrophs will germinate and subsequent treatment of the spore suspension with a suitable compound would kill the germinated prototrophic spores but leave the ungerminated auxotrophic spores unharmed. The auxotrophic spores may then be isolated by washing, to remove the inhibitor, and cultured on supplemented medium. Ganju and Iyengar (1968) developed a technique of this type using sodium pentachlorophenate against the spores of *Penicillium chrysogenum*, *Streptomyces aureofaciens*, *S. olivaceus* and *Bacillus subtilis*.

The mechanical separation of auxotrophic and prototrophic spores of filamentous organisms has been achieved by the 'filtration enrichment method' (Catcheside, 1954). Liquid minimal medium is inoculated with mutated spores and shaken for a few hours, during which time the prototrophs will germinate but the auxotrophs will not. The suspension may then be filtered through a suitable medium, such as sintered glass, which will tend to retain the germinated spores resulting in a concentration of auxotrophic spores in the filtrate.

The visual identification of auxotrophs is based on the alternating exposure of suspected colonies to supplemented and minimal media. Colonies which grow on supplemented media, but not on minimal, are auxotrophic. The alternating exposure of colonies to supplemented and minimal medium has been achieved by replica plating (Lederberg and Lederberg, 1952). The technique consists of allowing the survivors of a mutation treatment to develop colonies on petri dishes of supplemented medium and then transferring a portion of each colony to minimal medium. The transfer process may be 'mechanized' by using some form of replicator. For bacteria the replicator is a sterile velvet pad attached to a circular support and replication is achieved by inverting the petri dish on to the pad, thus leaving an imprint of the colonies on the pad which may be used to inoculate new plates by pressing the plates on to the pad. It may be possible to replicate fungal and streptomycete cultures using a velvet pad, but, if unsatisfactory results are obtained, a steel pin replicator may be more appropriate.

Visual identification of auxotrophs has also been achieved by the so-called 'sandwich technique'. The survivors of a mutation treatment are seeded in a layer of minimal agar in a petri dish and covered with a layer of sterile minimal agar. The plate is incubated for 1 or 2 days and the colonies developed are marked on the base of the plate, after which a layer of supplemented agar is poured over the surface. The colonies which then appear after a further incubation period are auxotrophic, as they were unable to grow on the minimal medium.

EXAMPLES OF THE USE OF AUXOTROPHS FOR THE PRODUCTION OF PRIMARY METABOLITES

Many auxotrophic mutants have been produced from *C. glutamicum* for the synthesis of both amino acids and nucleotide related compounds. *C. glutamicum* is a biotin-requiring organism which will produce glutamic acid under biotin-limited conditions but it is important to remember that mutants of this organism, employed for the production of other amino acids, must be supplied with levels of biotin optimum for growth. Biotin-limited conditions will result in these mutants producing glutamate and not the desired amino acid.

Auxotrophic mutants of *C. glutamicum* have been used for the production of lysine. The control of the production of the aspartate family of amino acids in *C. glutamicum* is shown in Fig. 3.14. Aspartokinase, the